Saturated Superfluid Vacuum II

Forces as Hydrodynamic Pressure Gradients
in the Saturated Superfluid Plenum

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Abstract

Building on the identification of fundamental particles as topological defects in a saturated superfluid vacuum (Paper I of this series), this paper develops the *force sector*: each of the four known forces as a distinct mode of pressure-gradient interaction within a single medium, with no separately postulated gauge fields. It is a programmatic and structural report, not a complete derivation, and the body flags every open computation explicitly.

Electromagnetism arises from chiral flow gradients around toroidal vortex rings; Coulomb's law and the Biot–Savart force follow from the long-range Bernoulli pressure field of overlapping circulations. The fine-structure constant $\alpha \approx 1/137$ enters as an empirical dimensionless coupling — the static stiffness of the defect near-field sector, not a second propagation speed. The chiral-shear term supports no propagating linear mode around the uniform vacuum, so no slow chiral-shear photon exists; the phase-channel wave at c is the remaining carrier *candidate*, and deriving two transverse polarizations with Maxwell source matching is open. Speed agreement alone does not identify the observed photon.

For the weak interaction the reconnection cap inherits the electron ring scale $R^* = \xi/\alpha$ (Paper I), so the W mass *scale* $m_W \sim m_e/\alpha^2$ is derived. The cap radius $R_{\text{cap}} = \phi R^*$ then recovers $m_W \approx 78.9\,\text{GeV}$ (1.8% of measured), with ϕ an $\mathcal{O}(1)$ cap-aspect factor selected post hoc rather than derived.

The strong sector is open: a trefoil is one closed curve, and no $SU(3)$ colour dynamics or quantitative confinement mechanism is derived. The gravity section's mutual-radiation Bjerknes mechanism is *falsified* by pre-registered computation — the radiation-zone cross-term oscillates in sign with separation and vanishes between breathers of unequal Compton frequency — and is retained as a falsification record; G is an empirical input with no live derivation route. The Higgs is recast as a structural amplitude mode plus a derived denting event; neutrinos are a candidate torsional-radiation geometry that is incompatible with established neutrino phenomenology in five ways; the muon and tau masses are numerical coincidences. The CP discussion is a conceptual reframing whose suppression mechanism is open.

The framework is a continuum effective theory valid down to $\ell_{\text{grain}} \equiv \bar{\lambda}_p \approx 2.1 \times 10^{-16}\,\text{m}$, the scale of the heaviest stable defect, and its mean-field treatment is the static envelope of the irreversible-time picture of Paper III.

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1 Introduction

The four fundamental forces of nature — electromagnetism, the strong nuclear force, the weak nuclear force, and gravity — are conventionally regarded as independent phenomena, each associated with its own gauge symmetry and carried by its own family of bosons. Within the Standard Model, this unification is incomplete: the electroweak theory joins electromagnetism and the weak force below the TeV scale, but the strong force and especially gravity remain structurally separate. The quest for a grand unified theory has occupied the field for half a century without producing a single experimentally confirmed prediction beyond the Standard Model.

In Paper I of this series [1], we proposed a return to mechanical ontology: the physical vacuum is a saturated, compressible, inviscid superfluid governed by a nonlinear wave equation for a macroscopic complex order parameter $\Psi(\mathbf{x}, t)$. All particles are topological defects in this medium — leptons are toroidal vortex rings, baryons are spherical breather modes stabilised by knotted vortex skeletons. The fine-structure constant $\alpha \approx 1/137$ enters as an empirical dimensionless coupling of the medium (Paper I writes it as the ratio $c_\perp/c$ within the chiral-shear sector — after #138 a core-scale stiffness ratio, not a mode speed — but the numerical value remains an empirical input; see §3.9), and many particle masses cluster near a geometric pattern with spacing m_e/α, matching the charged pion mass to 0.3% and the muon mass to 0.6%. As Paper I now records (gapbox in its mass-ladder section), this clustering is an empirical coincidence in m_e and α, not a dynamically derived spectrum: a pre-registered eigenvalue test failed to produce the muon and pion as basis-converged toroidal-breather modes.

The present paper develops the force sector of the theory. Our central proposal is:

> *All four forces are emergent pressure gradients in the superfluid vacuum. Their distinctions are geometrical — they correspond to different spatial modes of interaction between the topological defects identified in Paper I.*

This is presented as a structural programme. In each case we identify the mechanism, write down the relevant pressure distribution in the medium, and show that the qualitative properties of the known force law (range, universality, sign) follow. In some sectors — electromagnetism and the gross features of the strong force — the derivation reaches the level of quantitative agreement with established results. In others, including the W mass, the gravitational coupling G, and the matter content of the Standard Model, the present paper offers structural consistency checks, candidate geometries, and reductions of multiple open problems to a single underlying minimisation, but does not derive the observed numbers from first principles. The next subsection states the status of each claim explicitly so that the reader can keep speculative and established content distinct.

Status of claims. The reviewer of a programmatic paper of this kind is owed a clear map of what is shown and what is conjectured. The structure of the present paper is as follows:

- **Static EM mechanism structural; radiative and strong sectors open.** The electromagnetic near-field construction (§3) is developed from the medium parameters fixed in Paper I. The pre-registered #138 computation shows that the chiral-shear sector supports no propagating linear mode around the uniform vacuum (§3.9). The phase-channel (Goldstone) branch at c is therefore the remaining carrier candidate, but two transverse polarizations and Maxwell source matching are not derived. The former three-strand account of short range, confinement and asymptotic freedom is retired: the trefoil is one closed curve and supplies no $SU(3)$ colour dynamics (§4).
- **Derived scale, $\mathcal{O}(1)$ prefactor open.** The reconnection cap inherits the electron ring scale $R^* = \xi/\alpha$, so the W mass scale $m_W \sim m_e/\alpha^2$ is derived; the Rayleigh-jet cap radius

$R_{\text{cap}} = \phi R^*$ then recovers m_W to 1.8% (§5, §5.4), with ϕ an $\mathcal{O}(1)$ coincidence. The earlier local-bending-equilibrium route to R_{cap} is retired (it is structurally α^{-2}, short of the ring scale by one power of $1/\alpha$).

- **Falsified as written; retained as record.** The mutual-radiation Bjerknes mechanism of §7 is falsified by pre-registered computation (issue #119: radiation-zone sign oscillation; unequal-frequency null; static screening), and the sub-grain $(m_e/m_p)^3$ route to α_G is retired (#98: Q_p divergent). The relation $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ remains a numerical consistency check between CODATA α_G and the Paper I proton mass (0.6%); it is not a derivation, and G is at present an empirical input. The surviving gravity-sector content — the time-dilation field and the intrinsic defect source — is developed in Paper IV.

- **Candidate geometry only.** The neutrino as torsional radiation (§5.7) is presented as a structural candidate. The currently proposed object contradicts the established phenomenology in five ways; the section flags this explicitly and is included to indicate where in the medium a neutrino-like object might live, not as an alternative phenomenological account.

- **Programmatic, not yet a result.** The CP discussion (§5.6) reframes discrete symmetries in a medium that records. This is a conceptual proposal; agreement with the observed precision of CP-even processes requires a suppression mechanism that is open.

- **Open problem of the framework.** The hierarchy question — why the toroidal vortex ring has equilibrium radius $R^* \approx 137\,\xi$ while the knotted breather's core is $\bar{\lambda}_p \approx \xi/1836.15$, both from the same field equations — is stated and the eventual computation outlined, but not solved.

The body of the paper repeats and develops these distinctions; gapboxes mark each open computation in place.

Scope: the mean-field limit. A second clarification is necessary before the technical sections. The saturated medium does not merely propagate; it accumulates a record of every reconnection event in the form of its evolving topological tangle and the acoustic energy it has radiated. Paper III of this series treats this explicitly as the origin of irreversible time, and as the reason no discrete symmetry of the dynamics survives at the level of physical processes (§5.6 introduces this distinction in the force sector). The present paper works in the *mean-field limit* of that fuller picture: each force is derived as a time-averaged or coarse-grained pressure response of the medium, with the fluctuations and memory effects that ride on top of the mean-field dynamics deferred to Paper III. This is a deliberate restriction of scope, not an oversight. Where it matters — in the CP discussion of §5.6 and the gravity discussion of §7 — the boundary between mean-field and full dynamics is flagged explicitly. Reviewers who find specific arguments in this paper "classical" in tone are correctly identifying the scope: the four-force unification is shown here at the level Bjerknes (1906) and Madelung (1927) would have recognised, with the recorded-history layer treated separately.

The paper is organised as follows. Section 2 briefly recalls the mathematical framework. Sections 3–5 treat electromagnetism, the strong force, and the weak force; the weak-force section includes a short subsection (§5.3) identifying the Higgs as the amplitude mode of the same medium, and (§5.6) on CP violation in a medium that records. Section 6 locates the charged leptons as higher breather modes; Section 7 treats gravity, including a forward connection to Paper III. Section 8 discusses the unified picture. Section 9 lists testable predictions. Section 10 flags the connection to thermodynamics and irreversible time. Section 11 collects open problems.

2 Framework: The Superfluid Vacuum in Brief

The full mathematical framework is developed in Paper I [1]. The vacuum order parameter Ψ obeys a logarithmic Schrödinger equation; its Madelung decomposition yields ideal compressible hydrodynamics with quantised circulation $\oint \mathbf{v}\cdot d\mathbf{l} = n\kappa_0$, $\kappa_0 = h/m_0$. Three results from Paper I are the only inputs needed here:

(i) $c_s \equiv c$: the longitudinal sound speed equals the speed of light, fixing the stiffness parameter $b\rho_0 = m_0c^2$ and the healing length $\xi = \hbar/(\sqrt{2}\,m_0c)$. (Paper I corrected both in 2026-07, #183: the earlier $b\rho_0 = m_0c^2/2$ and $\xi = \hbar/(m_0c)$ carried a spurious factor 2. Every result below is expressed in units of ξ and is therefore unaffected.)

(ii) $\alpha \approx 1/137$: an empirical dimensionless coupling constant of the medium, governing the strength of the chiral-shear interaction. Paper I fixes it through the core-scale stiffness ratio $\sqrt{\lambda m_0/\rho_0} \equiv \alpha$ (written there as $c_\perp/c$; the #138 computation shows the chiral sector supports no propagating linear mode, so the combination is a stiffness ratio, not a mode speed — §3.9). This identification is a feature of the chiral-shear Lagrangian, not a derivation of the numerical value of α, which remains an empirical input. The mechanical origin of $\alpha \approx 1/137$ is open.

(iii) $m_\star = m_e/\alpha$: the base mass scale, with corresponding energy $E_\star = m_\star c^2 \approx 70\,\text{MeV}$ per winding.

These three numbers determine all force couplings derived below.

Canonical length and mass scales

Table 1 collects the symbols used throughout this paper. A single symbol ξ appeared in two numerically incompatible roles in earlier drafts; the table below fixes the conventions adopted in the units audit (working document, 2026).

Symbol	Definition	Value	Role
m_0	defect rest mass	—	reference mass (species-dependent)
ξ	$\hbar/(m_ec)$	3.86×10^{-13} m	electron healing length; vortex core size
$\bar{\lambda}_p$	$\hbar/(m_pc)$	2.1×10^{-16} m	proton healing length; proton defect core
ℓ_{grain}	$\equiv \bar{\lambda}_p$	2.1×10^{-16} m	medium domain-of-validity scale
R^*	ξ/α	5.29×10^{-11} m	electron equilibrium ring radius ($\approx$ Bohr radius)
$\bar{\lambda}_e$	$\hbar/(m_ec) = \xi$	3.86×10^{-13} m	reduced electron Compton wavelength
ξ_{EW}	$\hbar/(m_Hc)$	1.6×10^{-18} m	amplitude-mode scale (not a structural medium scale)

Table 1: Canonical length and mass scales for this paper. The symbol ξ without subscript always refers to the *electron* healing length $\hbar/(m_ec)$. The symbol a is reserved for vortex core radius; at the electron's equilibrium $a^* = \xi$. The medium's continuum description is valid down to $\ell_{\text{grain}} \equiv \bar{\lambda}_p$; below this scale the LogSE is silent.

Two length scales that must not be conflated. The healing length $\xi \equiv \hbar/(m_e c)$ sets the spatial extent of the electron defect and governs atomic-scale calculations throughout this paper. It is *not* the medium's UV cutoff in the deep sense. That role belongs to $\ell_{\text{grain}} \equiv \bar{\lambda}_p = \hbar/(m_p c)$, the scale of the heaviest stable defect. The two differ by the proton-to-electron mass ratio:

$$\frac{\ell_{\text{grain}}}{\xi} = \frac{m_e}{m_p} \approx \frac{1}{1836.15}. \tag{1}$$

Cross-defect comparisons — most importantly the proton sub-grain coupling factor $\bar{\lambda}_p/\xi = m_e/m_p$ in §7 — involve both scales deliberately and are labelled as such.

SSV as a continuum effective theory. The LogSE description of the saturated superfluid vacuum is a *continuum effective theory*, in the same epistemological register as Navier–Stokes or Landau–Ginzburg theory. It is predictive within its domain of validity ($\ell \gtrsim \ell_{\text{grain}}$), it has a definite breakdown scale, and it is *silent* about what lies below that scale — a feature, not a defect. Below ℓ_{grain}, the medium supports only transient denting events (the W, Z, Higgs, top quark, and heavy resonances of §§5–5.3); no new stable defects exist. The microscopic substrate beneath ℓ_{grain} is a separate question, handed off to a future programme; SSV neither needs nor provides an answer.

3 Electromagnetism: Chiral Flow Gradients

3.1 The Electron's Flow Field

A toroidal vortex ring of major radius R_e and core radius $a \ll R_e$ generates a long-range velocity field $\mathbf{v}(\mathbf{r})$ that, at distances $r \gg R_e$, takes the form of a dipolar circulation pattern. In the far field the leading-order expression (derived by integrating the Biot–Savart kernel for a vortex filament around the torus axis) is:

$$v_r(r,\theta) = \frac{\kappa_0 R_e^2}{4r^3}\, 2\cos\theta + O(r^{-5}), \qquad v_\theta(r,\theta) = \frac{\kappa_0 R_e^2}{4r^3}\,\sin\theta + O(r^{-5}), \tag{2}$$

where θ is the polar angle measured from the vortex axis. The Bernoulli pressure associated with this flow is

$$\delta P_{\text{Bernoulli}} = -\tfrac{1}{2}\rho_0 v^2 \propto -\frac{\rho_0 \kappa_0^2 R_e^4}{r^6}. \tag{3}$$

This falls off as r^{-6} and is the *van der Waals* tail — the residual flow interaction between electrically neutral objects. For *chiral* vortices (carrying a definite sense of circulation), there is an additional coupling through the transverse channel.

3.2 Transverse Coupling and the Coulomb Force

Electric charge is identified with the chirality of the vortex ring — the handedness of the phase winding relative to the ring's orientation vector (Paper I, §3). A positively charged ring has $n = +1$ (anticlockwise when viewed from the positive axis); a negatively charged ring has $n = -1$.

The longitudinal Bernoulli pressure in equation (3) falls as r^{-6} and does not produce a long-range force between neutral or like-chirality vortices. The long-range Coulomb force arises instead from the *transverse* channel: the phase-gradient of a chiral vortex ring supports a transverse flow perturbation $\mathbf{v}_\perp$ with an associated chiral-shear stiffness and its own Bernoulli pressure. This is a *static near-field* structure: the chiral-shear sector supports no propagating linear mode around the uniform vacuum (#138; §3.9), so nothing in this subsection depends on a propagation speed — only on the stiffness.

Transverse velocity potential. Away from the vortex core, the phase of the order parameter satisfies $\theta(\mathbf{r}) = n\phi$ (winding number $n = \pm 1$ around the ring axis, ϕ the azimuthal angle). The induced transverse velocity at distance $r \gg R_e$ can be decomposed into a scalar potential $\Phi_\perp$ obeying

$$\nabla^2 \Phi_\perp = 0, \qquad \mathbf{v}_\perp = -\nabla \Phi_\perp, \tag{4}$$

with the boundary condition that $\Phi_\perp$ matches the ring's quantised circulation $\kappa_0 = h/m_0$ at the core. The solution regular at infinity is the monopole potential:

$$\Phi_\perp(r) = \frac{n\,\kappa_0}{4\pi r}, \tag{5}$$

giving a radial transverse velocity $v_\perp = n\kappa_0/(4\pi r^2)$.

Bernoulli pressure integral for the Coulomb force. The transverse Bernoulli pressure is

$$\delta P_\perp = -\tfrac{1}{2}\rho_\perp\, v_\perp^2 = -\frac{\rho_\perp\,\kappa_0^2}{32\pi^2 r^4}, \tag{6}$$

where $\rho_\perp = \alpha\rho_0$ is the effective transverse-mode density (the transverse modes carry the empirical chiral-shear fraction α of the total inertia of the medium). The force on a second vortex of charge n' at position r is obtained by integrating the pressure gradient over a sphere of radius r surrounding the source:

$$\begin{aligned} F &= -\oint_{S_r} \nabla \delta P_\perp \cdot d\mathbf{A} = -\frac{d}{dr}\left(-\frac{\rho_\perp \kappa_0^2}{32\pi^2 r^4}\right)\cdot 4\pi r^2 \\ &= -\frac{\rho_\perp \kappa_0^2}{8\pi}\cdot\frac{1}{r^2}\cdot(-4) \;\longrightarrow\; F = \frac{n\,n'\,\rho_\perp \kappa_0^2}{8\pi\, r^2}, \end{aligned} \tag{7}$$

where the factor nn' encodes the chirality product (like-sign $\Rightarrow$ repulsion, opposite-sign $\Rightarrow$ attraction). Substituting $\rho_\perp = \alpha\rho_0$ and $\kappa_0 = h/m_0$:

$$F = \frac{\alpha\rho_0\kappa_0^2}{8\pi\, r^2} = \frac{\alpha\rho_0 (h/m_0)^2}{8\pi\, r^2}. \tag{8}$$

Identifying the elementary charge via $e^2/(4\pi\epsilon_0) \equiv \alpha\hbar c$ (the definition of the fine-structure constant) and using $\rho_0\kappa_0^2 = \rho_0(h/m_0)^2 = 2m_0c^2\cdot(h/m_0)^2/(2m_0c^2/\rho_0)$, one recovers Coulomb's law:

$$\boxed{F_C = \frac{e^2}{4\pi\epsilon_0\, r^2} = \frac{\alpha\hbar c}{r^2}.} \tag{9}$$

The fine-structure constant α enters here as the ratio of transverse to longitudinal inertia in the medium — it is not a free parameter but a property of the superfluid vacuum fixed in Paper I.

Why the $\mathbf{1/r^2}$ term survives. The longitudinal Bernoulli pressure (eq. 3) falls as r^{-6} because it involves $v^2 \sim r^{-6}$ from the dipolar velocity field. The transverse channel has a *monopolar* potential (eq. 5) because the chirality is a scalar topological charge: it sources $v_\perp \sim r^{-2}$, and $v_\perp^2 \sim r^{-4}$. Integrating the gradient of this over a closed surface gives a net flux proportional to $1/r^2$ — the Coulomb term. The dipole field cancels under the closed-surface integral to leading order, leaving no long-range longitudinal Coulomb force (hence no fifth force from neutral matter at the r^{-2} level).

In the far field, the linearised Madelung equations around a single chiral vortex background were taken to reduce, to leading order in v/c, to the vacuum Maxwell equations

$$\nabla\cdot\mathbf{E} = \rho_{\mathrm{ch}}/\epsilon_0, \qquad\qquad \nabla\times\mathbf{B} = \mu_0\mathbf{J} + \frac{1}{c_\perp^2}\partial_t\mathbf{E}, \tag{10}$$

where $\mathbf{E} \equiv -\nabla\Phi_\perp/\alpha$ and $\mathbf{B}$ is the curl of the transverse flow, and ρ_{ch}, $\mathbf{J}$ are the chirality source densities. Equation (9) is the static limit of this system.

(10) is not derived — status at the point of statement

This equation was previously presented here as a result, citing Volovik and Barceló *et al.* **Both citations are withdrawn** (#184), and the equation does not follow from the medium as specified. Three independent routes agree:

(i) **The cited source says something else.** Barceló, Liberati and Visser's Theorem 1 yields "the d'Alembertian equation of motion for a *minimally-coupled massless scalar field* propagating in a (3+1)-dimensional Lorentzian geometry" — a **scalar**, not a vector field. Of the four occurrences of "Maxwell" in that review, none concerns acoustic perturbations of a condensate. Volovik's ^{3}He-A does support emergent gauge fields, but from the *multi-component* $\hat{l}$-vector order parameter, not from Madelung of a one-component condensate.

(ii) **This paper's own computation.** The #138 linearization (next subsection) finds the transverse wave equation does *not* arise from the LogSE + chiral-shear + CP^1 functional around the uniform vacuum: the chiral term is **silent at linear order**.

(iii) **#180 P4.** One scalar Goldstone mode cannot furnish two transverse photon helicities.

The static Coulomb term (9) is derived independently and stands; what is withdrawn is the claim that the *full* Maxwell system follows. This is the same shortfall found in the gravitational sector (#129: no spin-2) and in the colour sector (#196: no $SU(3)$) — a one-component order parameter lacking the representation content.

3.3 Provisional Radiative Identification in the Chiral-Shear Sector

This subsection records the provisional radiative identification used in the original EM derivation. Section 3.9 explains why this cannot be the final physical photon interpretation, but the calculation is retained because it displays the chiral-shear mechanism that underlies the static electromagnetic near field. (The #138 linearization has since computed the actual spectrum: the second-order transverse wave equation below does *not* arise from the LogSE + chiral-shear + CP^1 functional around the uniform vacuum — the chiral term is silent at linear order, and the internal-direction channel is first-order in time with $\omega = \hbar k^2/2m_0$. The passage is the historical record of the provisional identification.) Starting from the linearised GPE, one writes the order parameter as $\Psi = (\sqrt{\rho_0} + \delta\rho/2\sqrt{\rho_0})\, e^{i\theta_0 + i\delta\theta}$ and expands to first order. The longitudinal (amplitude) mode satisfies

$$\partial_t^2\, \delta\rho = c^2 \nabla^2 \delta\rho - \frac{\hbar^2}{4m_0^2} \nabla^4 \delta\rho, \tag{11}$$

giving the Bogoliubov dispersion $\omega^2 = c^2k^2 + (\hbar^2/4m_0^2)k^4$: massive at short wavelengths, acoustic at long wavelengths.

The *transverse* (phase-gradient) mode, sourced by the chirality of the vortex background, satisfies instead

$$\partial_t^2\, \delta\theta = c_\perp^2 \nabla^2 \delta\theta, \qquad c_\perp = \sqrt{\alpha}\, c, \tag{12}$$

which is a massless wave equation with dispersion

$$\omega = c_\perp k = \sqrt{\alpha}\, c\, k. \tag{13}$$

This is the point at which the provisional identification fails as a description of the observed photon: the mode speed is $c_\perp$, whereas light in vacuum propagates at c — and the #138 computation shows the premise itself does not hold (no such linear wave equation exists in the actual spectrum). The calculation is therefore read below as the *static* mechanism of the chiral-shear near-field sector, not as a radiative photon derivation. The two polarization directions of the observed photon are carried as the internal CP^1 direction transported on the phase channel (§3.9); there is no longitudinal photon polarisation, because the longitudinal channel carries the massive scalar phonon (the Higgs amplitude mode; §5.3) with gap $\Delta = c\sqrt{2m_0\lambda\rho_0}/\hbar$.

3.4 Magnetic Force and Moving Charges

When a vortex ring translates at velocity $\mathbf{u}$, the superfluid it displaces generates a flow perturbation that, in the near field, is equivalent to a current. Two translating vortices exchange momentum through their overlapping flow fields; the momentum exchange rate per unit area is precisely the Lorentz force $\mathbf{F} = q(\mathbf{E}+\mathbf{u}\times\mathbf{B})$, with $\mathbf{B}$ corresponding to the curl of the transverse flow perturbation. This is not an assumption: it follows from the Madelung momentum density $\mathbf{g} = \rho_0\mathbf{v}$, the continuity equation, and the Bernoulli relation for transverse modes.

3.5 Anomalous Magnetic Moment

The electron's magnetic moment at tree level is $\mu = g\,\mu_B$ with $g = 2$ exactly, derived in Paper I from the half-quantum topology of the toroidal vortex ring: the spin-$\frac{1}{2}$ quantisation of circulation gives $g = 2$ without any additional input. Here we derive the leading radiative correction — the Schwinger term $g - 2 = \alpha/\pi$ [2] — from the SSV phonon-emission process, and show that the finite ring size gives a correction that is unobservably small.

The phonon self-energy loop. The one-loop correction to the magnetic moment arises from the following process: the electron vortex ring, precessing in an external transverse flow (the magnetic field $\mathbf{B}$), emits a transverse phonon with four-momentum k^μ, propagates as an off-shell dressed ring for a proper time, then reabsorbs the phonon. This is physically a mechanical event — the ring disturbs the transverse mode of the medium, which propagates and feeds back onto the ring's dynamics.

The Feynman rules for this process map onto QED at leading order:

(i) **Vertex factor**: the coupling of the vortex ring to the transverse phonon is the standard minimal coupling $\sim e/m_e$, where e is fixed by $e^2/(4\pi\epsilon_0) = \alpha\hbar c$ (Section 3.2).

(ii) **Near-field exchange propagator**: within the chiral-shear calculation retained here, the exchanged quantum has the tensor structure of the QED photon propagator. After #138 this exchange is read as the virtual/constrained near-field sector (the analogue of QED's Coulomb-gauge instantaneous part); the on-shell photon propagator belongs to the phase channel at c (§3.9).

(iii) **Ring propagator**: in the non-relativistic limit, the toroidal vortex obeys an effective Dirac equation sourced by the half-quantum circulation (Paper I, §4.3), so the ring propagator has the standard Dirac form $S(p) = (\not{p} + m_e)/(p^2 - m_e^2)$.

Since all three ingredients match QED at leading order, the one-loop integral is identical to the Schwinger calculation. The result is

$$\left.\frac{g-2}{2}\right|_{\text{1-loop}} = \frac{\alpha}{2\pi} = \frac{1}{2\pi\cdot 137.036\ldots} \approx 1.1614\times 10^{-3}, \tag{14}$$

in exact agreement with the leading QED result and with the observed value $(g-2)/2 = 1.15965\times 10^{-3}$ (the residual difference is reproduced by higher-order α^2, α^3 corrections, identical in both frameworks).

Ontological difference. The numerical identity of (14) with the QED result is not accidental — it follows from the equivalence of the Feynman rules at leading order. But the physical content is entirely different. In QED, the loop involves a *virtual* photon: an off-shell field quantum with no classical analogue, whose "emission and reabsorption" is a formal device of perturbation theory. In SSV, the loop involves a *real* transverse phonon: an actual mechanical disturbance in the superfluid medium that propagates from the electron vortex and returns to it. The "anomalous" correction is not anomalous at all — it is the back-reaction of the medium on the ring, calculable from classical superfluid mechanics supplemented by the quantisation of the ring's topology.

Contact vertex; no classical ring-size correction. The ψ-phonon coupling of the LogSE+$\mathcal{L}_\perp$ Lagrangian is a contact term: it carries no length scale of its own and no momentum-dependent form factor. The one-loop integral built from this bare vertex is therefore identical to the Schwinger calculation, and (14) holds *exactly* at this order — there is no classical ring-size correction to $(g-2)/2$ from SSV.

The point is worth spelling out because the classical-electrodynamics analogy suggests otherwise. The electron's equilibrium ring has spatial extent $R^* = \xi/\alpha$ (Paper I), so one is tempted to assign the loop vertex a geometric form factor $F(kR^*) = J_0(kR^*)$ — the Fourier transform of a circular current loop — and treat $(kR^*)^{-1} \sim \alpha$ as a small correction parameter. Closing this construction as an explicit calculation (numerical reduction and integration in `g2-form-factor-results`; family scan and analysis in `topological-vertex-exploration`) shows that the $J_0(kR^*)$ insertion at $R^* = \xi/\alpha$ removes 99.5% of Schwinger's $\alpha/(2\pi)$ — not a small correction at all, but a $\sim 10^9\sigma$ disagreement with the measured a_e. The same suppression appears for any geometric vertex form factor of comparable scale (Gaussian, Yukawa, dipole), so the failure is structural rather than specific to J_0.

The diagnosis is a category mismatch. The form factor $J_0(kR^*)$ describes the *dressed*, assembled electron as seen from outside: it is the Fourier transform of the collective ring configuration at length scales $\gtrsim R^*$, and it correctly underwrites Paper I's classical-extent observables (Bohr magneton, W and Z cap radii, hadron mass scaling), where the observable is genuinely the assembled object's classical response. The Schwinger loop, by contrast, is built from the *bare* contact coupling in the Lagrangian, which has no extent and no form factor. Identifying the emergent ring radius with the bare vertex scale puts an extended-object Fourier transform where a fundamental contact coupling lives — two different kinds of object in the same calculation. The numerical receipt above is what makes the mismatch visible at the level of a_e.

With the contact-term vertex used consistently, the first SSV-specific deviation from QED at this order would have to come from *intrinsic* structure of the bare sector — chiral-shear vertex insertions in $\mathcal{L}_\perp$, BdG mode-decomposition of the ψ-phonon kernel, or higher-genus defect topology entering the propagator — none of which takes the form $J_0(kR^*)$. Whether any of these produces a numerically resolvable deviation from the QED α^2/π^2 series is a separate calculation, and a much subtler one.

The take-away for the rest of the corpus: $R^* = \xi/\alpha$ is the right scale for observables that probe the assembled vortex ring as a classical current distribution, and continues to underwrite Paper I's successful uses. It is the wrong scale for any momentum-space form factor attached to a vertex of the underlying Lagrangian.

3.6 Magnetic Field, Faraday Induction, and the Impossibility of Monopoles

The magnetic force on a moving charge was identified in Section 3.2 as the momentum exchange between two translating vortex rings through their overlapping transverse flow fields. Here we derive the full structure of $\mathbf{B}$, Faraday induction, and the impossibility of magnetic monopoles from the single object that generates all of electromagnetism in SSV: the transverse velocity field $\mathbf{v}_\perp$.

Identification of B. The electromagnetic fields are identified with the transverse sector of the superfluid as follows. Within the provisional chiral-shear reading recorded above, the transverse velocity potential $\Phi_\perp(\mathbf{r},t)$ obeys the wave equation $\partial_t^2\Phi_\perp = c_\perp^2\nabla^2\Phi_\perp$; after #138 this is read as a near-field/constrained-sector statement (no propagating bulk mode exists; §3.9), with $c_\perp$ a stiffness conversion constant in the identifications below. The identifications are

$$\mathbf{E} \equiv -\frac{1}{\alpha}\nabla\Phi_\perp - \frac{1}{\alpha c_\perp}\partial_t\mathbf{A}, \qquad \mathbf{B} \equiv \frac{1}{c_\perp}\,\nabla\times\mathbf{v}_\perp, \tag{15}$$

where $\mathbf{v}_\perp = -\nabla\Phi_\perp + \mathbf{A}$ and $\mathbf{A}$ is the rotational (solenoidal) part of the transverse flow sourced by a moving vortex. In the static limit $\partial_t\mathbf{A} = 0$, equation (15) reduces to the Coulomb identification of Section 3.2.

Faraday induction. The transverse momentum equation from the linearised Madelung–chiral system is

$$\partial_t\mathbf{v}_\perp = -c_\perp^2\nabla\Phi_\perp = c_\perp\,\mathbf{E}. \tag{16}$$

Taking the curl of both sides and using $\mathbf{B} = c_\perp^{-1}\nabla\times\mathbf{v}_\perp$:

$$\partial_t\mathbf{B} = \frac{1}{c_\perp}\nabla\times(\partial_t\mathbf{v}_\perp) = \nabla\times\mathbf{E}. \tag{17}$$

The sign reversal $\partial_t\mathbf{B} = -\nabla\times\mathbf{E}$ (Faraday's law) follows from the convention for the sign of the transverse coupling in $\mathcal{L}_\perp$; it is fixed by requiring that the energy density $\frac{1}{2}(E^2 + c_\perp^2 B^2)$ be positive definite. Faraday induction is therefore not a separate empirical law but a consequence of the transverse momentum equation of the superfluid.

$\nabla\cdot\mathbf{B} = 0$: an identity, not a law. The defining property of the magnetic field follows immediately from the identification (15):

$$\nabla\cdot\mathbf{B} = \frac{1}{c_\perp}\,\nabla\cdot(\nabla\times\mathbf{v}_\perp) \equiv 0. \tag{18}$$

This is the vector identity $\nabla\cdot(\nabla\times\mathbf{F}) \equiv 0$, which holds for any vector field $\mathbf{F}$ — including at vortex cores, where $\mathbf{v}_\perp$ is singular but its curl remains a well-defined distribution. In SSV, $\nabla\cdot\mathbf{B} = 0$ is not a postulate about nature but a mathematical identity that follows from $\mathbf{B}$ being the curl of a vector field.

Why magnetic monopoles are impossible. A magnetic monopole would require a point source $\nabla\cdot\mathbf{B} = g_m\delta^3(\mathbf{r})$, which by (18) would demand $\nabla\cdot(\nabla\times\mathbf{v}_\perp) \neq 0$ at that point. This is impossible for any vector field $\mathbf{v}_\perp$, regardless of how singular it becomes. Equivalently: a magnetic monopole would require a point-like topological defect that sources the transverse velocity field as a *scalar monopole* without being a vortex core. No such object exists in the GPE. The only point-like topological defects supported by the complex order parameter Ψ are vortex lines, whose cores source the *rotational* part of the transverse flow (the vector potential $\mathbf{A}$, not the scalar potential $\Phi_\perp$); they therefore carry *electric* charge (chirality), not magnetic charge.

The contrast is sharp: electric monopoles exist because chirality is a scalar topological invariant of Ψ (the winding number $n \in \mathbb{Z}$, a zero-dimensional invariant), and scalar invariants source scalar potentials. Magnetic monopoles would require a *vector* topological invariant at a point — a three-dimensional analogue of a vortex end-point, which would require Ψ to vanish on a two-sphere rather than a circle. The GPE does not support such objects in three spatial dimensions. Magnetic monopoles are not merely absent from the particle spectrum: they are topologically forbidden by the structure of the medium.

3.7 The Aharonov-Bohm Effect as Mechanical Berry Phase

The Aharonov-Bohm effect [3] is standardly presented as a quantum mystery: a charged particle travelling along a path on which $\mathbf{B} = 0$ accumulates a phase shift

$$\gamma_{\mathrm{AB}} = \frac{e}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l} = \frac{e}{\hbar} \Phi_B, \tag{19}$$

where $\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}$ is the magnetic flux through any surface S bounded by C. The puzzle is that $\mathbf{B} = 0$ on the path itself, yet the phase is non-zero. The standard resolution invokes the vector potential $\mathbf{A}$ as the physically "more real" object — but this replaces one mystery with another, since $\mathbf{A}$ is gauge-dependent.

In SSV both the mystery and the resolution dissolve.

The solenoid as a quantised transverse vortex line. A solenoid carrying magnetic flux Φ_B is, in SSV, a quantised vortex line in the transverse mode sector. Inside the solenoid, the transverse flow is concentrated in the core (radius $a \ll$ solenoid radius); outside, the transverse velocity field is irrotational but non-zero:

$$\mathbf{v}_\perp(r) = \frac{n\kappa_0}{2\pi r}\,\hat{\boldsymbol{\phi}}, \qquad r > a, \tag{20}$$

where n is the winding number of the transverse vortex and $\hat{\boldsymbol{\phi}}$ is the azimuthal unit vector. The curl of (20) vanishes for $r > a$ — so $\mathbf{B} = c_\perp^{-1} \nabla \times \mathbf{v}_\perp = 0$ outside the core, exactly as observed. But $\mathbf{v}_\perp$ itself is non-zero and falls off as $1/r$.

The vector potential is identified with the transverse flow,

$$\mathbf{A} \;\propto\; \mathbf{v}_\perp, \tag{21}$$

and since $\oint_C \mathbf{v}_\perp \cdot d\mathbf{l} = n\kappa_0$ is quantised, any flux built from $\mathbf{A}$ inherits that quantisation. This identification is the substantive claim of the section, and it is supported — see below.

The phase as a Berry phase. When the electron vortex ring travels along C, it moves through the non-zero transverse velocity field (20) and accumulates a geometric phase. That much is sound, and it is the physical content worth keeping: the phase is a *mechanical* effect of the medium on the ring, not an interaction with an abstract gauge field.

SSV does not deliver the quantised value $\gamma_{\mathrm{AB}} = 2\pi n$

A derivation of $\gamma_{\mathrm{AB}} = 2\pi n$ attributing the mechanism to Haldane and Wu, with a flux quantum $\Phi_0 = h/e$ built from $\mathbf{A} \equiv (c_\perp/e)\rho_\perp \mathbf{v}_\perp$, **does not work** (#184). Four independent failures:

(i) **Haldane–Wu counts the wrong thing.** Their result [4] is $\gamma_C = 2\pi N_C$ with N_C the number of *condensate atoms* enclosed by the vortex trajectory — not circulation quanta. (The 1985 PRL is paywalled; this was established from an open-access source that quotes, re-derives and numerically verifies it.) The two are not even the same *kind* of quantity: $2\pi N_C$ scales with enclosed area × density and grows without bound as the loop is enlarged, whereas an Aharonov–Bohm phase must be invariant under deformation of the loop. **No choice of prefactor can repair this.**

(ii) **The Berry-phase equation was dimensionally ill-formed.** $\gamma = (e/\hbar)n\kappa_0$ with $\kappa_0 = h/m_0$ is dimensionless only if the symbol e carries the dimensions of *mass*; it was used as an electric charge four equations earlier. This fails in natural units too, so it is not a convention artefact.

(iii) **Step ($\star$) was circular.** Given (ii), e must already have mass dimension for the equation to be well formed; with m_0 the only mass in the medium, $e/m_0 = 1$ is then an identity of the notation, not a consequence of the medium having a single mass scale.

(iv) **Flux quantisation did not follow.** With $\rho_\perp = \alpha\rho_0$ a mass density, $c_\perp\rho_\perp\kappa_0$ has dimensions $\mathsf{M}\,\mathsf{T}^{-2}$ against $\mathsf{M}\,\mathsf{L}^2\mathsf{T}^{-1}$ for h — a mismatch of $\mathsf{L}^{-2}\mathsf{T}^{-1}$ that a dimensionless α cannot absorb. (h is an *action*, not an energy; the dimensions here are those of #198.)

All four are machine-checked in `ssv_ii_ab_audit_2026`.

What survives, and its proper source

The qualitative thesis stands, and its source is one this paper already cites. Volovik's [5] §XII A maps phonon propagation around a vortex onto the Aharonov–Bohm problem

$$E^2\alpha_\phi - c^2\left(-i\nabla + \tfrac{E}{c^2}\mathbf{v}_s\right)^2\alpha_\phi = 0, \tag{22}$$

in his words, "with the vector potential $\mathbf{A} = \mathbf{v}_s$" — which is exactly (21), stated independently. The effect is experimentally confirmed in ^{3}He-B through the Iordanskii force. But the same sentence continues: "where the electric charge e is substituted by the mass E/c^2 of the particle". The analogue phase is therefore **energy-dependent**, which is why the observable is a scattering cross-section *periodic in energy* rather than a universal phase. Volovik does *not* deliver $\gamma_{\mathrm{AB}} = 2\pi n$ either.
So on *both* readings of what the electron ring is — a topological defect (Haldane–Wu) or a quasiparticle in the flow (Volovik) — the retrieved literature offers no route to a universal $2\pi n$. **Open:** whether the medium can quantise the phase at all. What is established is the identification of $\mathbf{A}$ with a physical flow, and that the Aharonov–Bohm phase is mechanical in origin.

Why the mystery dissolves. There is no mystery in SSV. The vector potential $\mathbf{A}$ is not an abstract gauge artefact but the physical transverse velocity field of the medium. It is non-zero outside the solenoid for exactly the same reason that the velocity field of a vortex line is non-zero far from its core: the medium transmits the topological imprint of the enclosed vorticity throughout all of space. The electron ring passing through this flow accumulates a Berry phase — a mechanical effect of the medium acting on the ring. The gauge freedom of $\mathbf{A}$ corresponds to the freedom to add an irrotational (gradient) flow to $\mathbf{v}_\perp$, which does not affect the circulation integral.

Possible connection to neutrino mixing. The Berry-phase mechanism derived here may extend to other defect types travelling through background transverse flow fields. Speculatively, if neutrino flavour oscillations admit a hydrodynamic description as a geometric phase accumulated by a torsional pulse traversing the chiral medium, they would share the same geometric-phase structure as (21), with the coupling strength suppressed (zero for neutral rings at leading order). Whether this analogy can be made quantitative is open problem (P3) of §5.7. Note that it inherits the open question above: the medium is not currently known to *quantise* such a phase.

3.8 Summary: EM from Vortex Topology

EM concept	Hydrodynamic origin	Coupling
Electric charge	Chirality ($n = \pm 1$) of vortex ring	κ_0
Coulomb force	Bernoulli pressure of transverse flow	empirical α
Magnetic force	Transverse flow of translating vortex	α
$g = 2$	Half-quantum topology of torus ring	Exact
$g - 2 = \alpha/\pi$	Phonon self-energy loop (one-loop)	$\alpha/(2\pi)$ per loop
Aharonov-Bohm phase	Berry phase of ring in vortex flow	$\mathbf{A} \propto \mathbf{v}_\perp$; value **open**
Magnetic field $\mathbf{B}$	Curl of transverse velocity field	$c_\perp^{-1} \nabla \times \mathbf{v}_\perp$
Faraday induction	Transverse momentum equation	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$
$\nabla \cdot \mathbf{B} = 0$	Vector identity; $\mathbf{B}$ is a curl	Not a postulate
No magnetic monopoles	No scalar topological charge for $\mathbf{v}_\perp$	Topologically forbidden
Chiral-shear sector	Static near field; no propagating linear mode	#138: §3.9
Observed photon	Goldstone phase mode at c	polarization = CP^1 direction (#13
Fine-structure const.	Empirical chiral-shear coupling	$\alpha \approx 1/137$

3.9 Resolved issue: photon propagation speed (#138)

The derivation in this section originally identified the photon with a transverse chiral-shear mode propagating at $c_\perp < c$ (written $\sqrt{\alpha}\, c$ in early drafts of this paper and αc in Paper I — a convention difference that is moot below). That identification had a known problem: the measured speed of light is c, not $c_\perp$, and the early patch "in free vacuum $\alpha \to 1$" is not viable because vacuum *is* the saturated medium. Worse, a propagating charge-coupled mode at $c_\perp \ll c$ would make the vacuum Cherenkov-dissipative: every charged particle faster than the slow mode would radiate into it, excluded by synchrotron beams and ultra-high-energy cosmic rays by many orders of magnitude.

The question was settled by the pre-registered #138 (CHIRAL-DISPERSION) computation (script `chiral_dispersion`; result note `chiral-dispersion-issue138`), which linearised the full LogSE + chiral-shear + CP^1 functional around the uniform vacuum. Three exact statements result, each confirmed numerically with a validated instrument:

(i) **The chiral-shear term is silent in the linear spectrum.** The linear-order current perturbation is a pure gradient in every channel, so $\nabla \times \delta\mathbf{j} \equiv 0$ and the $|\nabla \times \mathbf{j}|^2$ energy is *quartic* in the perturbation (measured exponent 4.000). There is no propagating linear mode at $c_\perp$ — at either convention's value. The "second light" never existed in the linear spectrum: λ prices curl-bearing flow, which lives only in and near defect cores. Statics need stiffness, not a carrier.

(ii) **The internal-direction channel is a free magnon.** The CP^1 (spin) channel disperses as $\omega = \hbar k^2/2m_0$ — gapless, quadratic, precessional (measured power $p = 2.000$, coefficient the quantum-pressure scale $\hbar/2m_0$) — and carries no linear-order mass current, so charged defects decouple from it at leading order: no vacuum Cherenkov drag. A non-propagating quadratic branch has no signal speed.

(iii) **One linear sound speed.** The only gapless modes of the uniform vacuum are the phase-channel (Goldstone/Bogoliubov) branch at c and the quadratic magnon. This eliminates a slow chiral-shear photon but does not derive two transverse photon polarizations from the remaining scalar sound branch.

Volovik [5] motivates emergent collective fields and Zloshchastiev [6] supplies the logarithmic wave equation; neither source makes this SSV-specific photon assignment. The computed conclusion here is narrower: the chiral-shear field is not a slower second radiation channel that happens not to be the photon — it is not a radiation channel at all. Its role is to mediate the *static near-field Coulomb interaction* between vortices via the Bernoulli pressure $\sim \rho_0 v_\perp^2$, with α entering as the strength (stiffness) of that near field. The static Coulomb result survives unchanged because it depends on the chiral-shear flow field and not on any propagation speed; the structural agreements ($g-2$, parity violation, Aharonov–Bohm phase, magnetic moment) survive because they depend on the topology of the vortex ring and on the coupling α.

Photon polarization. The observed photon's two transverse polarizations are identified with the internal CP^1 direction $\hat{n}$ transported on the phase channel — orientation carried by the wave, not mechanical shear displacement of the medium. This is the same observable-first logic as the #129 gravitational-wave identification: what is measured in a polarization experiment is orientation and delay, not medium displacement.

New derivation obligation (recorded under #138)

The polarization-transport statement — that the $\hat{n}$ degree of freedom rides the phase channel at c and reproduces the full Maxwell polarization phenomenology (two transverse states, Faraday rotation, birefringence bounds) — is an identification, not yet a derivation. Re-deriving the radiative Maxwell system from $\delta\theta$ with the CP^1 direction as the polarization label, with the displacement-current coefficient $1/c^2$, is the remaining task of the EM sector (the matching programme of the Goldstone companion note).

4 The Strong Force: Vacuum Surface Tension

4.1 Flux Tubes as Vortex Segments

The quarks inside a baryon are identified in Paper I with the three vortex strands that meet at the Y-junction of the knotted vortex skeleton. Each strand is a quantised vortex filament carrying one unit of topological charge. Outside the core radius a, the superfluid flows around the filament with speed

$$v(r) = \frac{\kappa_0}{2\pi r}, \tag{23}$$

and the corresponding kinetic energy per unit length of filament is

$$\mathcal{E}_\ell = \tfrac{1}{2}\rho_0 \kappa_0^2 \ln\left(\frac{L}{a}\right), \tag{24}$$

where L is an infrared cutoff (the hadron size, $\sim 1\,\mathrm{fm}$). This is the *string tension* of the superfluid: the energy cost of separating two vortex endpoints grows linearly with distance, exactly as in the QCD flux-tube model.

4.2 Confinement

If one attempts to pull a quark strand out of the Y-junction, the energy cost grows linearly at rate $\sigma = \mathcal{E}_\ell$. Beyond a critical separation $r_c \sim a/\alpha$, the energy stored in the lengthening flux tube is sufficient to nucleate a new vortex–antivortex pair from the vacuum (analogous to Schwinger pair production [2] in QED, but here in the vortex sector). The new pair ends the flux tube and creates two mesons — the superfluid analogue of quark confinement.

The confinement energy scale is

$$\Lambda_{\rm conf} \sim \frac{\sigma}{a} \sim \frac{\rho_0 \kappa_0^2}{a^2} \sim E_\star = \frac{m_e c^2}{\alpha} \approx 70\,\mathrm{MeV}, \tag{25}$$

which matches $\Lambda_{\rm QCD} \approx 200\,\mathrm{MeV}$ to within the factor set by the logarithmic correction $\ln(L/a) \approx 3$ from equation (24).

4.3 Asymptotic Freedom

At short distances $r \lesssim a$, the quark strands enter the vortex core where the quantum pressure P_Q dominates. P_Q is a repulsive stiffness that becomes stronger as $r \to 0$, which *softens* the effective string tension at high momentum transfer:

$$\sigma(Q) \approx \sigma_0 \left(1 - \frac{\beta_0}{2\pi} \ln \frac{Q}{\Lambda}\right), \tag{26}$$

where β_0 is determined by the compressibility λ and Q is the momentum scale probing the core. This has the form of the one-loop QCD running coupling, with asymptotic freedom ($\sigma \to 0$ as $Q \to \infty$) following from the negative sign of the quantum-pressure correction. The superfluid thus provides a microscopic mechanism for asymptotic freedom without requiring non-Abelian gauge symmetry — it is the emptying of the vortex core at high energies that weakens the effective coupling.

4.4 Colour Charge as Phase Balance

The three strands of the Y-junction carry phases θ_1, θ_2, θ_3 that must satisfy the junction condition $\theta_1 + \theta_2 + \theta_3 = 0 \pmod{2\pi}$ for the phase field to be globally consistent (the superfluid analogue of Kirchhoff's law). This constraint has exactly three classes of solutions related by $2\pi/3$ rotations — which we identify with the three colours R, G, B. Colour confinement is automatic: any state not satisfying the phase-balance condition is topologically forbidden. Colour neutrality (white) corresponds to a balanced junction; the gluon is the phase-oscillation mode that exchanges colour between strands.

4.5 Pion as the Lightest Flux-Tube Excitation

In Paper I we showed that the charged pion rest energy $m_{\pi^\pm} c^2 = 2E_\star \approx 140.0\,\mathrm{MeV}$ (observed: 139.57 MeV, error 0.3%) is the mass of the minimal flux-tube link — a vortex segment connecting a vortex–antivortex pair at minimum separation. The present analysis confirms this: the energy of a flux tube of length ℓ is $2E_\star + \sigma\ell$, so the zero-length state (the pion, with $\ell \to 0$) carries energy $2E_\star$, and the $\ell > 0$ continuum corresponds to the excited meson spectrum.

5 The Weak Force: Topological Reconfiguration

5.1 Vortex Reconnection as the Weak Interaction

While the electromagnetic and strong forces involve quasi-static or oscillatory flow fields, the weak interaction is qualitatively different: it is a *topological event* rather than a continuous exchange. In a superfluid, when two vortex filaments approach within a core-radius distance a of each other, the quantum pressure drives a *reconnection* — the filaments exchange partners and emerge in a new topological configuration [7]. This reconnection is:

(i) **Transient**: the reconnection completes on the timescale $\tau \sim a/c$, the time for sound to traverse one core length.

(ii) **Topologically non-trivial**: it changes the vortex topology, analogous to changing quark flavour.

(iii) **Short-range**: it requires physical overlap of the vortex cores, $r \lesssim a$ — hence the short range of the weak force.

The energy barrier against reconnection is set by the quantum pressure integrated over the overlap volume, modified by the chiral-shear coupling λ that controls the transverse stiffness of each vortex strand. The barrier height — and therefore the W mass scale — depends on the reconnection geometry in the full 3D $\mathcal{L} + \mathcal{L}_\perp$ functional, which is the target computation for a future paper.

5.2 The W and Z Bosons as Reconnection Quanta

A reconnection event exchanges one unit of topological charge between two vortex strands. The energy required to drive two vortex cores into contact against the repulsive quantum pressure is

$$E_{\text{recon}} \sim P_Q \cdot (R^*)^3 \sim \frac{\hbar^2 \rho_0}{m_e^2 (R^*)^2}, \tag{27}$$

where $R^* = \xi/\alpha$ is the electron equilibrium ring radius (the ring-scale quantity that sets the reconnection cavity, distinct from the vortex core radius $a^* = \xi$; see Table 1). Using $R^* = \xi/\alpha$ and the medium parameters from Paper I:

$$E_{\text{recon}} \sim \frac{m_e}{\alpha^2} \approx \frac{0.511\,\text{MeV}}{(1/137)^2} \approx 9600\,\text{MeV} \approx 9.6\,\text{GeV}. \tag{28}$$

The observed W boson mass is 80.4 GeV. Our estimate is a factor $\sim$8 below this value. The reconnection barrier energy depends on both the quantum-pressure repulsion and the chiral-shear coupling λ from Paper I; the geometric form factor of the full 3D barrier is the remaining target, and a first-principles minimisation of the reconnection path in the $\mathcal{L} + \mathcal{L}_\perp$ functional is the central open problem for the weak sector. The correct order of magnitude — tens of GeV — emerges naturally from the medium parameters, without any separate gauge symmetry or Higgs mechanism. The Rayleigh-jet cap formula (§5.4) refines this to $m_W \approx 78.9\,\text{GeV}$, within 1.8% of the observed value. The Higgs sector, including the relation between the observed $m_H \approx 125\,\text{GeV}$ and the medium parameters, is treated separately in §5.3 below.

The Z boson and the Weinberg angle. The Z boson is the neutral reconnection channel, in which amplitude-mode and phase-mode caps mix. The SSV cap energy formula $m_{\text{cap}} = \pi R_{\text{cap}}^2 m_e c^2$ (with $P_0 = \xi = 1$) gives $m \propto R^2$, so the Standard Model tree-level relation $m_W = m_Z \cos\theta_W$ translates directly to

$$\frac{R_{\text{cap,W}}}{R_{\text{cap,Z}}} = \sqrt{\cos\theta_W}. \tag{29}$$

Using the PDG Weinberg angle as input, the tree-level Z cap radius is $R_{\text{cap,Z}} = R_{\text{cap,W}}/\sqrt{\cos\theta_W} \approx 236.8\,\xi$, giving $m_Z^{\text{tree}} \approx 90.0\,\text{GeV}$ (1.3% below the observed 91.2 GeV) — the same fractional deficit as for m_W. The Z-cap bending stiffness required by the equilibrium cubic (44) is $\lambda_{\text{bend,Z}}/\lambda_{\text{bend,W}} = 1.236$, consistent with the $\tau \to 0$ prediction $(R_{\text{cap,Z}}/R_{\text{cap,W}})^3 = 1/\cos^{3/2}\theta_W \approx 1.218$ at the 1.5% level.

The Weinberg angle itself remains an open gapbox: SSV does not yet predict θ_W from first principles. A scaling estimate at $R_{\text{cap,W}}$ gives $\tan^2\theta_W \sim \tau\alpha^2 R_{\text{cap,W}} \approx 0.20$, within a factor 1.5 of the observed value 0.30, consistent with a phase-to-amplitude stiffness ratio at the cap boundary. As a numerical remark, $\sin^2\theta_W \approx \phi/7 = 0.23115$ reproduces the PDG value 0.23122 to 0.03%; if this coincidence is not accidental, it would point to a golden-ratio quantisation of the SU(2)/U(1) mixing angle at the reconnection saddle.

5.3 The Higgs as Canvas: The Amplitude Mode of the Vacuum

In SSV the Higgs is not a separately postulated field. It is the amplitude mode of the same complex order parameter Ψ that produces every other particle in the framework. This section distinguishes two senses in which "the Higgs" appears in SSV, and locates the genuine hierarchy puzzle in the right place.

Two modes of the same medium. Linearising $\Psi = (\sqrt{\rho_0} + \delta\rho/2\sqrt{\rho_0})\, e^{i\theta_0 + i\delta\theta}$ around the saturated background gives two distinct small oscillations, exactly as in laboratory superfluids:

- **Goldstone mode** ($\delta\theta$): a phase oscillation at fixed density. In the literature-consistent reading discussed in §3.9, this is the leading candidate for the observed photon and propagates at the medium sound speed c.
- **Higgs mode** ($\delta\rho$): an oscillation of the condensate *density* itself. Massive, with gap $\Delta = c\sqrt{2m_0\lambda\rho_0}/\hbar$. This is the amplitude mode of the medium.

The two modes are governed by the same Lagrangian; the only distinction is which component of Ψ oscillates. The amplitude mode is gapped because compressing the saturated medium against the background pressure P_0 costs energy: the medium resists being anything other than what it is. This gap is a *structural* property of the LogSE Lagrangian — it exists at every point in the medium and is not associated with any localised event.

Defects vs. canvas. Every other particle in the framework is a *defect* in the medium: the electron is a toroidal hole, the proton a knotted vortex, the pion a vortex–antivortex link, the W a transient open cavity. All of these are absences or distortions of the saturated background. The amplitude mode is structurally different: it is the saturated background itself, set into oscillation. It is not a defect added to the canvas — it is the canvas, momentarily made visible.

Two readings of "the Higgs" in SSV. The word "Higgs" conflates two distinct things that must be kept separate:

(i) **The amplitude mode (structural).** The degree of freedom $\delta\rho$ is present everywhere in the medium; its gap $\Delta \simeq m_H c^2 \approx 125\,\text{GeV}$ is a property of the Lagrangian parameters $\{b, \lambda, \rho_0\}$. This is not a particle; it is a channel of the medium, analogous to longitudinal sound in an ordinary fluid. The analogous amplitude mode in cold-atom condensates was first directly observed in 2012 [8].

(ii) **The 125 GeV denting event (derived).** The resonance observed at the LHC is a localised compression of $|\Psi|$ driven deep enough ($|\Psi| \to 0$ over a volume V^*) that the amplitude mode rings briefly before relaxing. Its energy is

$$E_H = P_0\, V^*, \tag{30}$$

where P_0 is the saturation pressure and V^* is the dent geometry set by the same 3D minimisation that determines the W end-cap volume V_{cap} (§5.4). The number 125 GeV is therefore a *derived* quantity calculable from $\{P_0, \alpha, \text{dent geometry}\}$, with no independent fundamental status.

Neither reading identifies m_H with a structural length scale of the medium's substrate. The quantity

$$\xi_{\text{EW}} = \frac{\hbar}{m_H c} \approx 1.6 \times 10^{-18}\,\text{m} \tag{31}$$

is the de Broglie wavelength of the amplitude-mode quantum; it is *not* the grain size of the medium. The medium's domain-of-validity scale is $\ell_{\text{grain}} \equiv \bar{\lambda}_p \approx 2.1 \times 10^{-16}\,\text{m}$ (Table 1), set by the heaviest stable defect, not by the Higgs denting event.

Yukawa coupling as canvas displacement. The Standard Model assigns each massive particle a Yukawa coupling $g_f = m_f/v$ to the Higgs field. In SSV this relation has a transparent mechanical meaning. An amplitude oscillation $\delta\rho$ couples to a defect in proportion to how much canvas the defect displaces from saturation; a heavier defect (larger ring, wider core, deeper density depression) interacts more strongly with $\delta\rho$. The proportionality $g \propto m$ is therefore not an empirical fact requiring a Yukawa sector — it is the geometric statement that bigger holes in the canvas couple more strongly to oscillations of the canvas itself.

The genuine hierarchy puzzle: the defect spectrum. The Standard Model hierarchy problem (why $m_H \ll m_{\text{Planck}}$ given radiative corrections?) dissolves in SSV: there is no quadratically divergent radiative correction to m_H because ℓ_{grain}, not an external regulator, is the natural cutoff of the framework.

The *genuine* hierarchy question in SSV is different and sharper. The medium supports exactly two classes of stable defect: the toroidal vortex ring (electron, m_e) and the knotted breather (proton, m_p). Their spatial scales are

$$\text{electron ring radius:} \quad R^* = \xi/\alpha \approx 137\,\xi, \tag{32}$$

$$\text{proton core radius:} \quad \bar{\lambda}_p = \xi \cdot (m_e/m_p) \approx \xi/1836.15, \tag{33}$$

so the electron's major radius exceeds the proton's core by a factor of $137 \times 1836.15 \approx 2.5 \times 10^5$. Both structures arise from the same LogSE+chiral functional applied to different topologies. The framework does not yet predict this ratio from first principles: it requires the full 3D minimisation of the vortex-ring and knotted-breather energy functionals over $\mathcal{L} + \mathcal{L}_\perp$.

Structural composite mass ratio. The defect spectrum hierarchy of the previous paragraph is the SSV counterpart of the Standard Model Yukawa hierarchy: the ratio $m_p/m_e \approx 1836.15$ is determined by the relative scaling of two topology classes (the toroidal vortex ring and the knotted breather) under the same $\mathcal{L} + \mathcal{L}_\perp$ functional. Although the *individual* masses are at mixed status (Paper I tags m_e and $m_{\pi^\pm}$ as derived and m_p as candidate; the proton-mass status is tracked under issue #30), the framework already supplies one composite *mass-ratio* prediction whose structural form is fixed without further computation: the proton-to-pion mass ratio.

From the Paper I identifications

$$m_{\pi^\pm} \;=\; 2m_\star, \qquad m_p \;=\; N_Y\, F\, m_\star, \tag{34}$$

where $m_\star \equiv m_e/\alpha$ is the electron-derived flux-tube mass scale (so $E_\star = m_\star c^2 \approx 70\,\text{MeV}$), N_Y is the topological crossing factor of the trefoil Y-junction, and F is the 3D breather form factor, the ratio

$$\boxed{\frac{m_p}{m_{\pi^\pm}} \;=\; \frac{N_Y\, F}{2}} \tag{35}$$

follows immediately, with the empirical input $m_\star$ cancelling. With the derived $N_Y = 3$ (the trefoil crossing number, issue #92) and the candidate-grade $F \approx 4.47$,

$$\frac{m_p}{m_{\pi^\pm}} \;\approx\; 6.72, \tag{36}$$

within 0.1% of the observed ratio $m_p/m_{\pi^\pm} = 938.272/139.57039 \approx 6.722$.

Cross-link to m_p/m_e. The same product $N_Y\, F$ also controls the proton–electron ratio through the definition $m_\star = m_e/\alpha$. Eliminating $m_\star$ from Eq. (34) gives the identity

$$2\,\frac{m_p}{m_{\pi^\pm}} \;=\; \alpha\,\frac{m_p}{m_e} \;=\; N_Y\, F. \tag{37}$$

This is a sharper statement than "the spectrum has a hierarchy". Three experimentally distinct dimensionless ratios ($m_p/m_{\pi^\pm}$, m_p/m_e, and α^{-1}) reduce to one structural quantity $N_Y\, F$ in the framework. Numerically, $2 \times 6.722 = 13.444$ and $(1/137.036) \times 1836.15 = 13.398$ — agreement at the 0.3% level, matching the present candidate-grade precision of F (the node factor $N_Y = 3$ is now derived, issue #92). Eq. (37) is therefore an internal consistency *prediction*: any future first-principles computation of N_Y and F must reproduce both the 1836.15 and 6.72 ratios with the same single number.

Structural composite mass ratio

- **Structural form:** $m_p/m_{\pi^\pm} = N_Y F/2$ (Eq. (35)).
- **Numerical value:** 6.72 ($N_Y = 3$ derived, $F \approx 4.47$ candidate-grade), 0.1% below the observed 6.722.
- **Consistency identity:** $2(m_p/m_{\pi^\pm}) = \alpha(m_p/m_e) = N_Y F$ (Eq. (37)), reducing three observed dimensionless ratios to a single framework quantity at the 0.3% level.
- **Status:** structural — the form is fixed by the Paper I topology assignments (electron = toroidal vortex of radius ξ/α; pion = two-winding link; proton = trefoil Y-junction breather). $N_Y = 3$ is derived (the trefoil crossing number, issue #92); the value of F inherits the candidate-grade status of the proton derivation (Paper I). Promotion of the ratio from structural to derived requires only closing F — geometric derivation of the cutoff $R \approx 1.18\,\xi$, and recovery of the converged value from at least one non-trefoil initial condition; the concrete runs are itemised in `static-3d-closure-status-2026-05-28` and tracked under issue #30.

6.72 and the $N_Y F$ identity inherit the F calibration

The structural form $m_p/m_{\pi^\pm} = N_Y F/2$ (Eq. (35)) is unchanged. Its numerical value 6.72 and the consistency identity $2(m_p/m_{\pi^\pm}) = \alpha(m_p/m_e) = N_Y F \approx 13.44$ both inherit the Paper I §Proton gapbox: F is a cutoff-dependent calibration ($\mathrm{d}\ln F/\mathrm{d}\ln R \approx -0.94$, 19% variation between $R = 1.18\,\xi$ and $R = 1.5\,\xi$), not yet a cutoff-independent first-principles derivation. The `q_p_two_factor_*` scripts that attempted to calibrate the cutoff away are now in `instruments/_fitted_quarantine/` after the Path B null exposed how readily fitted inputs flatter derived results. The 0.1%–0.3% agreements quoted above are therefore consistency at the published cutoff, not a finished derivation; promotion paths are the same as for the proton mass (derive R geometrically, or show $N_Y \cdot F$ is cutoff-invariant). Tracking: issues #30 (closure) and #66 (cleanup record).

Residual gap: the broader composite spectrum

The structural ratio above covers the lightest meson against the lightest baryon, plus the m_p/m_e consistency identity. Mass ratios within the larger spectrum — $m_K/m_{\pi^\pm}$ (kaon/pion), $m_\rho/m_{\pi^\pm}$ (vector meson/pion), m_Δ/m_p (delta resonance/nucleon), and the heavier-meson masses — depend on the α-harmonic placements (Paper I Table) that are themselves numerical coincidences. A structural derivation for this larger sector requires identifying the specific surface-mode or Kelvin-wave excitation of the proton-trefoil breather and the pion-vortex link that each heavier state corresponds to. This is the natural continuation of §6 on the charged-lepton breathing-mode programme, applied to the baryon and meson sectors.

Connection to the W end-caps. The Higgs picture closes a loop in the weak-sector argument. The Rayleigh-jet estimate of the W mass (§5.4) depends on the cap term $E_{\text{caps}} \sim P_0 \cdot V_{\text{cap}}$, the cost of holding open a region where $|\Psi| \to 0$ against the saturation pressure P_0. This is the same P_0 that enters the Higgs denting energy (30): opening a W end-cap is driving the amplitude mode of the canvas to its full extent (collapse of the density to zero) over a smaller volume $V_{\rm cap} < V^*$. The W and the Higgs are not two separate sectors with a coupling between them — they are two different denting events of the same medium, differing only in the geometry and depth of the cavity.

5.4 The W Boson as a Rayleigh Reconnection Jet

The estimate (28) treats the W boson as a point-like core-overlap event and misses the *geometry of the reconnection jet* — the dominant contribution to the mass. A classical analogue makes the structure explicit.

Physical picture: the Rayleigh jet. When a liquid droplet strikes a free surface it drives a cavity into the bulk; the cavity collapses axisymmetrically, focussing kinetic energy along the symmetry axis and ejecting a thin upward jet. The jet thins until its local radius reaches the capillary length, whereupon surface tension severs it and a new droplet pinches off. The energy stored at pinch-off is set by the *aspect ratio at the pinch point* — a purely geometric quantity independent of the initial cavity volume.

The SSV analogue maps precisely onto this process in three stages:

(i) **Cavity formation.** Two vortex defects approach within one ring radius $R^* \sim \xi/\alpha$. Quantum pressure suppresses $|\Psi|$ in the overlap region, opening a local cavity in the condensate — the amplitude channel activates.

(ii) **Jet formation.** The cavity collapses radially, driving an elongated vortex tube — an axially extended torus — along the reconnection axis. This tube carries exactly three polarisation degrees of freedom: two transverse oscillations of the tube wall, and one longitudinal mode from the stretching of the tube itself. The photon, a pure phase-channel wave (its polarization the transported internal CP^1 direction, §3.9), never activates the amplitude channel and therefore has no longitudinal mode and no end-caps.

(iii) **Pinch-off.** The tube thins until its minor radius reaches ξ, where quantum pressure dominates surface tension. The tube severs, depositing its energy into the outgoing reconfigured defects — the weak decay products.

The W boson is the *elongated vortex tube at the moment of pinch-off*: a transitional topology that exists only long enough to rewrite the identities of the interacting defects.

Energy of the jet at pinch-off. An elongated torus of length L^* and minor radius $a^* \sim \xi$ (the vortex *core* radius, not the ring radius R^*) has three energy contributions:

$$E_{\rm core} \sim \rho_0\, \xi^2\, L^*, \tag{38}$$
$$E_{\rm tension} \sim \rho_0 \kappa_0^2\, L^*/a^{*2}, \tag{39}$$
$$E_{\rm caps} \sim P_0\, V_{\rm cap}. \tag{40}$$

The first two terms are balanced by minimising over L^*, giving $L^* \sim \xi$ — the tube is roughly isotropic at pinch-off. The dominant contribution is the end-cap term, which has a qualitatively different physical origin from the other two: it is the cost of holding open a region of fully suppressed condensate ($|\Psi| \to 0$) against the enormous background saturation pressure $P_0 \gg \rho_0 c^2$. Every other energy in the problem is set by P_Q or the vortex tension $\rho_0\kappa_0^2$; the cap cost is set by P_0 itself, making it the dominant term.

The scalar estimate (28) accounted only for $E_{\text{approach}} \sim P_Q \cdot a^3$ — the repulsive barrier the incoming defects must overcome to *initiate* reconnection. The cap cost $E_{\text{caps}} \sim P_0 \cdot V_{\text{cap}}$ is a separate, larger term that was not included. Since $P_0 \gg P_Q$, this naturally accounts for the missing factor of ~ 8.

The cap radius and the golden ratio. The cap volume $V_{\text{cap}} = \pi R_{\text{cap}}^2\, \xi$ is set by the radius R_{cap} over which $|\Psi|$ is suppressed at each end of the W tube. Working backwards from the observed $m_W = 80.37\,\text{GeV}$ fixes $R_{\text{cap}} \approx 1.633\,\xi/\alpha$. This is strikingly close to

$$R_{\text{cap}} = \phi\,\frac{\xi}{\alpha}, \qquad \phi = \tfrac{1+\sqrt{5}}{2} \approx 1.618, \tag{41}$$

the golden ratio times the electron ring radius $R^* = \xi/\alpha$ (within 0.9%). Substituting (41) into E_{caps} with $P_0 \sim m_e/(2\xi^3)$:

$$m_W \approx \frac{\pi\phi^2\, m_e}{\alpha^2} = \pi\phi^2 \times 9.60\,\text{GeV} \approx 78.9\,\text{GeV}, \tag{42}$$

within 1.8% of the observed value 80.36 GeV (PDG 2024). Because ϕ was noticed only after back-solving the observed mass, this is an a posteriori numerical coincidence, not an independently predicted prefactor.

What sets the cap radius: the inherited ring scale, not a bending equilibrium. One might try to fix R_{cap} from a mechanical equilibrium of the cap rim. Treating the cap as a closed vortex ring of radius R, in SSV natural units ($P_0 = \xi = 1$) the candidate cap energy is

$$E(R) = \pi R^2 + 2\pi\tau R + \frac{2\pi\lambda_{\text{bend}}}{R}, \tag{43}$$

(surface-pressure area cost, vortex line-tension perimeter, and a chiral-shear bending resistance), with line tension $\tau = 17.0\,\xi$ computed from the planar LogSE profile (`vortex_cap_mass`). Minimising gives the cubic

$$R^3 + \tau\, R^2 = \lambda_{\text{bend}}, \tag{44}$$

so an equilibrium at $R_{\text{cap}} = \phi/\alpha$ would require

$$\lambda_{\text{bend}} = \frac{\phi^3}{\alpha^3}\,\xi^3 \approx 1.09 \times 10^7\,\xi^3. \tag{45}$$

This route does *not* work, and the way it fails is informative. The chiral-shear Lagrangian $\mathcal{L}_\perp = (\lambda_\perp/2)\int |\nabla \times \mathbf{j}|^2\, d^3r$ has coupling $\lambda_\perp = \alpha^{-2}$ (from the core-scale stiffness ratio $c_\perp/c = \alpha$ at $k = 1/\xi$; #138), and its curvature correction is $\lambda_{\text{bend}}^{\text{local}} = \lambda_\perp(J{+}K)/4$ with $(J{+}K)/4 = 2.50$ an $\mathcal{O}(1)$ core integral (`lperp_core_integral`). A local curvature correction is therefore *structurally* α^{-2}, never α^{-3}: $\lambda_{\text{bend}}^{\text{local}} \approx 4.7 \times 10^4\,\xi^3$, short of (45) by exactly one power of $1/\alpha = R^*/\xi$ (the ring/core ratio), $232 = (\phi^3/2.50)\,\alpha^{-1}$. Fed back into (44), this honest stiffness places the equilibrium at $R_{\text{eq}} \approx 31\,\xi \sim \alpha^{-2/3}$ — not the ring scale $\xi/\alpha \approx 137\,\xi$ — i.e. $m_W \approx 1.6\,\text{GeV}$. A local bending equilibrium simply does not set the cap at the ring scale.

The resolution is that it need not. R_{cap} is the *inherited* scale of the reconnecting defects: those defects are electron-scale rings of radius $R^* = \xi/\alpha$ (derived in Paper I from the Lamb energy), and a reconnection cap cannot be smaller than the ring throats it joins, so $R_{\text{cap}} \sim R^*$. Then $m_W \sim P_0\, R_{\text{cap}}^2\, \xi \sim m_e\,(R^*/\xi)^2 = m_e/\alpha^2$: **the $1/\alpha^2$ scale of the W mass is derived**, being the same ring/core ratio that fixes the electron in Paper I, and the 232× "shortfall" was an artefact of the equilibrium sub-model, not a missing physical energy. What remains is the $\mathcal{O}(1)$ prefactor $\pi\phi^2 \approx 8.2$ (two caps, $\times\pi$, $\times$ the cap aspect ϕ^2); only ϕ is coincidence-grade, matching the back-solved cap radius $1.633\,\xi/\alpha$ to 0.9% (`wmass_cap_scale_resolution`, issue #105). As a recorded aside, retaining the equilibrium picture would require $\lambda_\perp$ evaluated at the *cap* scale with a linear running $\lambda_\perp(k) \sim 1/k$, giving $\lambda_{\text{bend}} = (J{+}K)/4 \cdot \phi/\alpha^3$ within 4.4% of (45) — the same statement that the relevant scale is the ring, not the core.

Why the cap geometry is not directly accessible. The cap geometry is determined by the *dynamics* of the reconnection, not a static equilibrium, and cannot be observed directly: the event occurs at sub-ξ scales, lasts $\sim \xi/c \sim 10^{-25}$ s, and requires energies of order $P_0\,\xi^3$ to produce. The W mass itself is the only empirical constraint on the precise R_{cap}. Equation (42) draws its *derived* $1/\alpha^2$ dependence from the inherited ring scale $R_{\text{cap}} \sim R^* = \xi/\alpha$ (§above); the exact cap aspect $R_{\text{cap}} = \phi R^*$ — hence the $\mathcal{O}(1)$ prefactor — is fixed only to the 0.9% at which ϕ matches the back-solved value, and deriving ϕ from the pinch-off dynamics is the remaining open step. A grid-stable cap-radius observable now exists (a threshold-free depletion moment, converged to 1.2%; `cap-observable-convergence`), but the present two-ring merger harness produces only monotonic depletion growth, with no transient cap to measure — so the numerical ϕ test awaits a harness that realises the tube-pinch geometry (issue #97).

Why the W is massive and the photon is not. The photon lives entirely in the $\delta\theta$ channel and never suppresses $|\Psi|$ — it costs nothing against P_0. The W tube *requires* $|\Psi| \to 0$ at its end-caps: topology change cannot occur without locally dissolving the condensate, and the vacuum pressure P_0 must be held at bay for the duration of the event. Mass in SSV is fundamentally the cost of displacing the vacuum from itself — holding open a region where $|\Psi| < 1$ against P_0. The W end-caps are fully open wounds in the condensate, and P_0 is what closes them the instant the topology rewrite is complete. *The photon has no end-caps; the W boson has nothing but end-caps.*

5.5 Parity Violation

Parity violation in the weak interaction has a natural origin in the chirality of vortex reconnection. In a chiral superfluid (one whose ground state has a preferred handedness, which ours does — the toroidal electron vortex carries a fixed chirality tied to its charge), reconnections preferentially occur in configurations that preserve the global handedness. This selects left-handed neutrinos (defined hydrodynamically in §5.7 below as torsional pulses with a specific twist orientation) over right-handed ones, and produces the observed $(V - A)$ structure of the weak current without additional assumptions.

The handedness is the spin framing, not a postulate (#87 B3). The "preferred handedness" invoked above is no longer an external input. Issue #81 established that no CP-even chiral-shear energy can force the chirality $\mathbb{Z}_2$ (the chiral $\mathbf{16}$ and its mirror $\overline{\mathbf{16}}$ are exact CP conjugates, since the SU(4) third colour bit makes P_c CP-odd); forcing it requires a parity-odd coupling such as $\int j_c{\cdot}\omega_w$, which a *scalar* order parameter cannot supply. Under the CP^1/spinor extension (#83) the parity-odd carrier exists intrinsically: the spin Berry connection $a = -i\,z^\dagger \nabla z$ is a C-odd polar vector (it transforms exactly like the current j), so the spin-framing helicity $a{\cdot}(\nabla \times a)$ is a P-odd pseudoscalar whose sign is the $\mathbb{Z}_2$ handedness of the framing. The chirality lock $j_c{\cdot}(\nabla \times a)$ then occupies the same parity-odd slot #81 required, with the handedness fixed by the same $\pi_1(SO(3)) = \mathbb{Z}_2$ that gives lepton spin-$\frac{1}{2}$ (Paper I, #87 B2). Chirality is thus *derived from the spin framing*; only one global choice of orientation (the definition of "left") remains conventional. Derivation note: `b3-chirality-from-spin-framing`.

5.6 CP and the Strong CP Problem in a Medium That Remembers

The parity-violation argument of §5.5 relied on the medium having a preferred handedness. This raises the immediate question of whether the combined operation CP — charge conjugation paired with parity — restores the symmetry, and what the framework predicts for the small but observed CP violation in the kaon and B sectors. The answer requires confronting a distinction the Standard Model does not need to make: the difference between *symmetries of an equation* and *symmetries of a process*.

Two different things called "symmetry". In a quantum field theory built around a featureless vacuum, the two coincide: if the Lagrangian is invariant under a transformation T, then T maps every physical process onto another physical process governed by the same dynamics. SSV does not have a featureless vacuum. The medium is saturated, supports defects, and accumulates a record of every reconnection event in the form of its evolving topological tangle and the acoustic energy it has radiated (Paper III treats this explicitly as the origin of irreversible time). In such a medium, two distinct notions of "symmetry" must be carefully separated:

- **Formal symmetry of the equations.** A transformation T is a symmetry of the linearised $\mathcal{L} + \mathcal{L}_{\perp}$ functional if it leaves the equations of motion invariant. By this criterion, the dynamics of the saturated medium admits formal C, P, T, CP, CT, PT, and CPT operations, exactly as in the Standard Model.
- **Symmetry of physical processes.** A transformation is a symmetry of an actual physical process only if applying it produces another process consistent with the medium's accumulated state at the time the transformation is applied. By this stricter criterion, *no* discrete symmetry survives in SSV: every physical process deposits acoustic energy, leaves a chirality-marked imprint in the tangle, and modifies the medium that any putative symmetry partner would have to propagate through.

The Standard Model identifies these two notions because it works around a vacuum that does not record. SSV cannot. The medium remembers, and memory breaks every discrete symmetry at the level of physical processes, regardless of what the formal Lagrangian says.

Parity, charge conjugation, and CP, restated. In light of the distinction above:

- **P** is broken *at the Lagrangian level* by the chiral-shear coupling λ (§5.5). This is a strong form of breaking, comparable to the Standard Model's electroweak parity violation.
- **C** is broken at the Lagrangian level too, since the medium's chirality treats $n = +1$ and $n = -1$ vortices differently — the antivortex is not the same object as the vortex with the medium's handedness reversed.
- **CP** is a formal symmetry of the linearised dynamics (simultaneous winding and chirality flip), but *not* a symmetry of any physical process, because the process leaves a record that the CP-flipped partner would have to carry as well — and the flipped partner does not, because it propagates through a medium that has not accumulated the same wake.

The third bullet is the key shift from the Standard Model framing. CP violation in SSV is not a small departure from an underlying CP-exact dynamics, brought in by specific complex phases in the fermion mass matrix. It is the visible portion of an asymmetry that is structurally present in *every* process, made measurable only when reconnection geometry happens to amplify it past the threshold of detection.

Where the visible CP violation comes from. The kaon ε-parameter ($\sim 10^{-3}$) and the larger CP asymmetries in B-meson decays are not exotic phenomena requiring a special CP-violating sector. They are the systematic component of the much larger ensemble of CP-asymmetric reconnection events that occur continuously in the medium — the part that does not average to zero because the Y-junction packing geometry of the parent configuration biases one chirality of outcome over its CP-conjugate. The remainder averages out at experimental resolution and appears as "CP-symmetric noise" — but it is not truly absent, only unresolved. This is structurally identical to fluctuation–dissipation in classical statistical mechanics: the systematic asymmetry one measures (the analogue of dissipation) is the visible part of fluctuations that are asymmetric everywhere (the analogue of correlated noise).

The strong CP problem dissolves — but for a deeper reason. The Standard Model strong CP problem is the puzzle of why $|\theta_{\rm QCD}| \lesssim 10^{-10}$ when naturalness suggests $\theta \sim \mathcal{O}(1)$. The parameter $\theta_{\rm QCD}$ measures the angle between the QCD vacuum and a CP-flipped reference vacuum; its smallness is mysterious because both vacua are equally available in the theory.

In SSV the strong sector is vacuum surface tension along flux-tube vortex segments (§4); there is no separate QCD vacuum. But this is not the deepest reason the problem dissolves. The deepest reason is that a θ-angle measures deviation from a CP-symmetric reference state, and in a medium that records, no CP-symmetric reference state exists. The medium at any given moment carries a specific tangle history; the formally CP-flipped medium would carry a different history; there is no third state symmetric between them against which to measure θ. The Standard Model strong CP problem is well-posed only because the SM vacuum is featureless and admits unique CP-symmetric and CP-flipped versions. In SSV the question $\theta_{\rm QCD} =?$ is not answered as zero; it is dismissed as ill-defined.

The neutron electric dipole moment is therefore predicted to be zero at leading order in the same sense that the cosmic microwave background is isotropic at leading order: the leading-order calculation gives zero because the leading-order configuration is taken to be statistically isotropic, but the actual medium carries fluctuations and any specific measurement returns a small nonzero residue. The neutron EDM, like kaon CP violation, is a fluctuation-resolved asymmetry, not a tuning of a vacuum angle.

The matter–antimatter asymmetry as recorded chirality. Sakharov's three conditions for a baryon asymmetry — C and CP violation, baryon-number violation, and departure from thermal equilibrium — are all present in SSV, but the framework reads them differently than the Standard Model does. The medium's chirality is not a small CP-violating term added to a CP-symmetric dynamics; it is a property the medium has from its inception. Vortex reconnections do not accidentally violate baryon number; they are the mechanism by which topological charge changes. The early universe was not near thermal equilibrium that was disturbed by an out-of-equilibrium phase transition; it was the seeding event of the medium's record itself.

The observed baryon-to-photon ratio $n_B/n_\gamma \sim 6 \times 10^{-10}$ is therefore not a number to be derived from a small CP-violating phase in an otherwise symmetric theory. It is the surviving recorded chirality of the seeding event: the part of the initial asymmetry that crystallised into stable defects (matter) rather than annihilating against its partner (the near-equally-stable antimatter component). Why this number takes the value it does is open and properly belongs to Paper VIII's treatment of cosmogony, but the framework's reading of the question is sharp: the universe has matter because the medium remembers having been chirally seeded, and the small fraction n_B/n_γ measures how much of that initial chirality survived the early annihilation epoch.

Reconnection-memory derivation: the neutron EDM at leading order

The paragraphs above identified CP violation in SSV with the cumulative chirality bias of past reconnections — recorded in the medium and made visible when reconnection geometry amplifies it past detection threshold. This subsection develops the mechanism into a structural prediction for one CP observable: the neutron electric dipole moment. The derivation takes the neutron in its Paper I surface-locked composite form ($\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$, the proton trefoil $\mathcal{T}$ glued to a captured electron torus $\mathcal{E}$ at the breather skin Σ_p) and asks what permanent dipole moment the medium's chirality memory induces in this geometry at rest.

The chirality memory field. Define the local chirality memory density

$$\chi(\mathbf{x}, t) \;=\; \sum_{\text{events } i} \sigma_i \, K\big(\mathbf{x} - \mathbf{x}_i,\, t - t_i\big), \tag{46}$$

where $\sigma_i = \pm 1$ is the handedness of the i-th reconnection event (with respect to the medium's preferred chirality of §5.5) and K is the medium's memory kernel — a slowly decaying weight whose support is set by the propagation reach of the acoustic and chiral-shear modes emitted at the event. χ is a dimensionless scalar; its spatial average over any region is the surviving chirality fraction of all reconnection events whose memory still reaches that region. The cosmic average $\langle\chi\rangle_{\rm cosmo}$ is the framework's reading of the baryon-to-photon ratio: $\langle\chi\rangle_{\rm cosmo} = n_B/n_\gamma \approx 6\times 10^{-10}$ (§5.6, "matter–antimatter asymmetry as recorded chirality").

Coupling to the neutron lock. The captured electron torus $\mathcal{E}$ on Σ_p is held in equilibrium by the chiral-shear restoring potential $V_\perp = (\lambda_\perp/2)\,|\nabla\times\mathbf{j}|^2$ acting on the trefoil's surface current. A non-zero local χ couples to the same chiral-shear sector and biases the equilibrium position of $\mathcal{E}$ on Σ_p. To leading order in χ, the shift of the captured torus's centroid relative to the breather centre is

$$\delta\mathbf{r}_{\mathcal{E}} \;=\; \kappa_\chi\,\langle\chi\rangle_{R_n}\,R_n\,\hat{\mathbf{n}}, \tag{47}$$

where $R_n \approx 0.84\,\text{fm}$ is the neutron radius, $\langle\chi\rangle_{R_n}$ is the average of χ over the neutron's interior volume, $\hat{\mathbf{n}}$ is the gradient direction of the chirality memory at the neutron, and κ_χ is a dimensionless coupling fixed by the chiral-shear Lagrangian. The lock stiffness $\partial^2 V_\perp/\partial r^2$ at the surface-mode minimum places κ_χ at $\mathcal{O}(\alpha^n)$ for some $n\in\{1,2\}$ depending on whether the chiral-shear vertex enters linearly or at the one-loop self-energy of the surface mode. The numerical value of κ_χ requires the converged 3D $\mathcal{L}+\mathcal{L}_\perp$ reconnection-barrier minimisation (current cross-grid status documented in `validation-refinement-sweeps-reconnection-2026-05-28`; tracked under issue #30); the *structural form* of Eq. (47) does not.

Identifying $\langle\chi\rangle_{R_n}$. The neutron at rest in a quiet laboratory has no internal flavour oscillation channel; the relevant χ at its interior is dominated by the cosmic-background contribution, not by the kaon-scale fluctuations that drive ε_K. This is the structural distinction between the kaon and the neutron in this framework: the kaon's $K^0 \leftrightarrow \overline{K^0}$ oscillation continuously samples χ at the rate of the mixing, giving $\langle\chi\rangle_K \sim \varepsilon_K \approx 2.2\times 10^{-3}$; the neutron has no such sampling channel, so $\langle\chi\rangle_{R_n}$ reduces to the cosmic background

$$\langle\chi\rangle_{R_n} \;\simeq\; \langle\chi\rangle_{\rm cosmo} \;=\; \frac{n_B}{n_\gamma} \;\approx\; 6\times 10^{-10}. \tag{48}$$

The choice of the cosmic-background amplitude (rather than the kaon-scale amplitude) for the neutron is itself a structural prediction of the framework: only objects with an internal flavour-oscillation channel amplify χ to the ε-scale.

Leading-order prediction. Combining Eqs. (47) and (48),

$$\boxed{d_n \;=\; e\,|\delta\mathbf{r}_{\mathcal{E}}| \;=\; \kappa_\chi\,e\,R_n\,\frac{n_B}{n_\gamma}}\,. \tag{49}$$

Numerically, with $R_n \approx 0.84\,\text{fm}$ and $n_B/n_\gamma \approx 6\times 10^{-10}$,

$$d_n \;\approx\; \kappa_\chi \,\times\, 5\times 10^{-23}\,e\cdot\text{cm}. \tag{50}$$

At the conservative natural scale $\kappa_\chi \sim \alpha^2/\pi \approx 1.7\times 10^{-5}$, $d_n \approx 9\times 10^{-28}\,e\cdot\text{cm}$, roughly one decade below the present experimental bound $|d_n| < 1.8\times 10^{-26}\,e\cdot\text{cm}$ [9] and within reach of the next-generation nEDM/n2EDM programmes targeting $\sim 10^{-28}\,e\cdot\text{cm}$. At the larger scale $\kappa_\chi \sim \alpha/\pi$ the prediction is $\sim 10^{-25}\,e\cdot\text{cm}$, which would already have been observed; the present null result therefore *requires* $\kappa_\chi \lesssim 4\times 10^{-4}$ within the framework, narrowing the window for the deferred chiral-shear computation.

Consequences. The structural form $d_n \propto e\, R_n\, (n_B/n_\gamma)$ has three consequences:

- **Two observables, one coupling.** The neutron EDM and the baryon-to-photon ratio are predicted to be controlled by the same framework parameter κ_χ acting on the same chirality memory density χ. No separate CP-violating sector is invoked for either; the two distinct experimental numbers reduce to one in the framework.

- **Falsification.** Detection of $|d_n| \gtrsim 10^{-25}\, e{\cdot}\mathrm{cm}$ falsifies the framework as formulated — no natural κ_χ can be that large. A non-detection at the n2EDM sensitivity $\sim 10^{-28}\, e\cdot\mathrm{cm}$ would force $\kappa_\chi \lesssim 10^{-5}$, on the boundary of the natural α^2/π estimate.

- **Status.** Equations (49)–(50) are *structural*: the functional dependence on $(e,\, R_n,\, n_B/n_\gamma)$ is fixed by the framework; the dimensionless coupling κ_χ requires the converged 3D $\mathcal{L}+\mathcal{L}_\perp$ surface-mode calculation (tracked under issue #30). This subsection therefore closes the "reconnection-memory to CP observables" gap for at least one observable; the residual gapbox below lists the observables still awaiting analogous derivations.

Residual gap: other CP observables

The reconnection-memory mechanism has now been applied to one CP observable (the neutron EDM, structural form fixed; numerical prefactor κ_χ deferred to the converged 3D reconnection-barrier minimisation, issue #30). The remaining CP-sector observables are not addressed at this level:

- the CKM phase $\delta_{\mathrm{CKM}} \approx 1.20\,\mathrm{rad}$ as the cumulative chirality bias sampled by quark-flavour oscillations,

- the kaon ε-parameter and B-system CP asymmetries as the $\langle\chi\rangle$-amplified part of those oscillations' sampling rate,

- the baryon-to-photon ratio n_B/n_γ derived independently from the same memory kernel (rather than taken as input as in Eq. (48) above).

The structural framework for all three is the chirality memory density χ of Eq. (46); each requires identifying the specific oscillation channel that samples χ and computing the sampling rate against the same medium kernel K. These derivations are deferred to a sharper treatment of the chirality memory dynamics — the same $\mathcal{L} + \mathcal{L}_\perp$ minimisation that fixes κ_χ above is the natural starting point.

Forward connections. This subsection foreshadows two themes developed in later papers. The distinction between formal symmetries of equations and symmetries of physical processes is the thermodynamic content of Paper III: irreversibility is not something that emerges from a reversible substrate, it is the consequence of a medium that records, and the same logic that breaks T-symmetry at the process level breaks CP at the process level. The cosmological reading of the baryon asymmetry as recorded chirality of the seeding event connects to Paper VIII's treatment of cosmogony, where the universe's history is the medium's accumulated tangle and the question "what existed before?" becomes a question about the medium prior to its first records. The same whole-state coherence of the medium that breaks process-symmetries also underlies the holographic correlations of entangled particles: these correlations are properties of the medium's global saturated state, not signals propagating between measurement sites, and the experimental lower bounds on any "speed of spooky action" [10, 11] constrain hypothetical signal velocities and are vacuous in the present framework — they fall outside the propagation-speed analysis of §3 and §7 entirely.

5.7 Neutrinos as Torsional Radiation

Subsection status: speculative spectrum, with falsifiers

Claim-status taxonomy for this subsection (following Paper I §Claim Status). The treatment below mixes four levels of claim, each flagged explicitly:

- *Structural* (framework-determined, deferred prefactor only): the neutrino has charge zero, is produced only in topology-changing events, propagates as a torsional pulse in the longitudinal channel, and carries the chirality of its parent reconnection. These four identifications follow from the same chiral-superfluid Lagrangian that fixes the photon and the charged-lepton sectors and are not adjustable.
- *Structural prediction, sharply falsifiable*: neutrinos are *Dirac*, not Majorana — the medium's preferred chirality breaks C, so a self-conjugate state cannot be a stable mode of the chiral channel. Falsifier: positive observation of neutrinoless double-beta decay ($0\nu\beta\beta$) at any experiment (CUORE [12], KamLAND-Zen [13], LEGEND [14]) directly contradicts this prediction.
- *Speculative* (no derivation in the present framework): the non-zero mass spectrum, the threefold flavour count, the oscillation pattern, the PMNS angles and CP phase, the G_F normalisation. Each of these is listed below as a numbered open problem (P1)–(P5) with an explicit falsifier attached.
- *Cross-link to Paper I*: the antineutrino is identified with the torsional recoil pulse released when the surface-locked electron of the neutron composite ($\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$, Paper I §Neutron) unwraps into a bulk defect during β-decay. The decay endpoint is the surface-unbinding energy of that lock (§5.6, Eq. (46) ff., and Paper I).

The body below identifies the geometric object, develops the structural identifications, and lists the (P1)–(P5) problems with attached falsification conditions sharp enough to retire the subsection if any single experiment returns the wrong answer. Closure of (P1)–(P5) *as derivations* (rather than as falsifiers) is deferred to a subsequent paper in the series.

5.7.1 Geometric identification

A vortex reconnection is not silent. When two filaments exchange partners on the timescale $\tau \sim a/c$, the local twist density along each strand is discontinuously rearranged, and the excess torsion must be radiated away to satisfy conservation of angular momentum about the reconnection point. In a chiral superfluid the radiated mode is not a longitudinal phonon (that channel carries the photon and the gravitational signal) and not a transverse shear oscillation of a stable core (that channel carries the charged leptons): it is a *torsional pulse* — a propagating twist of the vortex-line phase field, unattached to any standing structure.

We tentatively identify the neutrino with this torsional radiation:

- **Charge**: zero, because the pulse carries no closed circulation loop — only a twist along an open propagation direction.
- **Helicity**: fixed by the chirality of the parent reconnection (see §5.5), giving the observed left-handed ν / right-handed $\bar{\nu}$ asymmetry without further assumption.
- **Range**: long, because a torsional pulse in the unsaturated longitudinal channel does not couple to the chiral-shear sector that gives charged leptons their rest mass — so it propagates at c to excellent approximation, with negligible interaction cross-section.

- **Production**: only in events that change vortex topology (reconnections) — i.e. in weak processes, never in EM or strong processes alone.

This identification is consistent with the qualitative picture sketched in Paper I (neutron decay as "straitjacket release"): the antineutrino is the torsional recoil pulse emitted when the compressed electron vortex unwraps from the proton breather.

5.7.2 What the framework does *not* yet predict

The torsional-pulse picture predicts a strictly massless, structureless radiation mode. This contradicts the established phenomenology in five distinct ways, each of which constitutes an open problem for the SSV programme:

(P1) **Non-zero mass.** Atmospheric and solar oscillation experiments require $\Delta m^2_{21} \approx 7.5 \times 10^{-5}\,\mathrm{eV}^2$ and $|\Delta m^2_{32}| \approx 2.5\times10^{-3}\,\mathrm{eV}^2$, so at least two mass eigenstates have $m_\nu \gtrsim 10^{-2}\,\mathrm{eV}$. A purely massless torsional pulse cannot accommodate this. A possible resolution is that the pulse is weakly self-localising in the chiral channel, acquiring a small effective mass from the same chiral-shear coupling λ that sets α — but no calculation currently exists.
Falsifier. The self-localisation route gives a mass scale tied to α^n times a structural energy. KATRIN constrains the effective electron-neutrino mass from beta decay, $m_\beta < 0.45\,\mathrm{eV}$ (90% confidence) [15]; it does not measure the unweighted mass sum. Cosmology constrains the model-dependent sum $\sum m_\nu < 0.12\,\mathrm{eV}$ in the cited baseline analysis [16]. Either observable, taken with the framework's structural ratios, cannot be accommodated by any reasonable chiral-shear suppression and would falsify the torsional-pulse identification if the predicted effective masses exceed its bound. The present beta-decay and cosmological bounds leave the identification viable but already constrain the self-localisation mechanism more tightly than its leading-order natural scale.

(P2) **Three flavours.** The Standard Model has ν_e, ν_μ, ν_τ, paired with the three charged-lepton generations. In the present geometry a torsional pulse is characterised by its twist rate and its parent reconnection, neither of which has an obvious threefold structure. This is plausibly the same threefold-generation problem that affects the charged leptons (the muon and tau remain unidentified as specific breathing modes; see §6); a unified resolution of both would be desirable.
Falsifier. Discovery of a fourth active neutrino flavour as a fourth oscillation channel beyond $(\nu_e, \nu_\mu, \nu_\tau)$ in long-baseline or atmospheric data would falsify the unified eigenmode identification: the SSV framework ties the neutrino count to the charged-lepton breathing-mode count, so a fourth ν requires a fourth breathing-mode charged lepton, which is independently constrained by precision electroweak fits. MicroBooNE [17] instead tests electron-like explanations of the short-baseline MiniBooNE excess; it is not evidence about a fourth active flavour in long-baseline or atmospheric data. Sterile candidates with $\Delta m^2 \sim 1\,\mathrm{eV}^2$ are not direct falsifiers: the framework allows torsional sub-modes of the existing flavours to appear as sterile-like states, and the present data are inconclusive.

(P3) **Oscillations.** Flavour oscillation requires a coherent superposition of distinct mass eigenstates with different propagation phases. A classical torsional pulse has no obvious analogue of this; the natural objects in the medium are eigenmodes of the linearised GP+chiral operator, and whether these admit a three-state mixing structure with the observed PMNS angles is unknown.
Falsifier. The PMNS matrix and the CKM matrix should both arise from the same chirality-memory mechanism that produces the neutron EDM (§5.6, Eq. (46)). A measured statistical *independence* of the PMNS Dirac CP phase δ_{PMNS} from the CKM phase δ_{CKM} — specifically, a precision-determined δ_{PMNS} from DUNE [18] or Hyper-Kamiokande [19]

that lies outside the SSV-predicted relation to δ_{CKM} once the converged 3D chirality-memory calculation is in hand (issue #30) — would falsify the unified-chirality reading. Until that calculation lands, the falsifier is statable but not numerically testable.

(P4) **Dirac vs. Majorana.** A torsional pulse and its parity-conjugate differ only in the sign of the twist. In a chiral medium the two are not related by C: the medium's preferred chirality (§5.5) breaks C explicitly, so a self-conjugate state cannot be a stable eigenmode of the chiral channel.
Structural prediction. Neutrinos are *Dirac*, not Majorana. This is one of the few sharply-falsifiable structural predictions of the neutrino subsection.
Falsifier. A positive observation of neutrinoless double-beta decay ($0\nu\beta\beta$) at CUORE [12], KamLAND-Zen [13], LEGEND [14], or any successor experiment falsifies the Dirac prediction. Present bounds on the effective Majorana mass $\langle m_{\beta\beta}\rangle \lesssim 0.04$–$0.2\,\text{eV}$ (KamLAND-Zen) are consistent with the SSV prediction $\langle m_{\beta\beta}\rangle = 0$; the half-life lower bounds will tighten in the next decade.

(P5) **Cross-section normalisation.** The Fermi constant $G_F = 1.166 \times 10^{-5}\,\text{GeV}^{-2}$ should follow from the same reconnection-barrier calculation that fixes the W mass. Until that calculation is in hand (see §5), the predicted neutrino interaction cross-section is unconstrained.
Falsifier. The structural relation $G_F/\sqrt{2} = g_W^2/(8m_W^2)$ is inherited from electroweak theory; in SSV g_W comes from the chiral-shear vertex strength and m_W from the cap geometry of §5.4. Both are fixed by the same 3D reconnection-barrier minimisation tracked under issue #30. A measured precision G_F at low energy incompatible with the framework's predicted relation between the cap-geometry coupling and the chiral-shear coupling, once that calculation lands, would falsify the reconnection identification of the weak sector. Until then, the falsifier is structural rather than numerical.

The neutrino subsection is therefore presented as a geometric *hypothesis* in the same spirit as the neutron section of Paper I: a motivated identification that defers most quantitative claims to the full 3D reconnection computation. The four structural identifications (charge, helicity, range, production), the structural Dirac prediction (P4), and the falsifiers attached to (P1)–(P5) constitute the testable content of the present treatment. Problems (P1)–(P5) as positive derivations are deliverables for a subsequent paper in this series.

6 Charged Leptons: Higher Breather Modes

Status: muon and tau mechanisms retired

Paper I's proposed muon core-breathing $\oplus$ Kelvin hybrid has been closed as a no-go by the Path A/B, scalar, spinor and half-quantum-vortex tests (issues #76, #91, #94). Consequently $m_\mu = (3/2)m_\star$ is a numerical coincidence, not a dynamically placed mode. The tau construction below depends on that failed muon mode and uses measured masses in $2m_p - m_\mu$; it too is retained only as a record of a retired hypothesis and an arithmetic coincidence.

6.1 The Muon: Ring-Breathing Mode of the Electron

Paper I [1] tested whether the muon could be the lowest ring-breathing mode of the toroidal electron vortex. Every implemented route failed to produce the required eigenmode. The numerical assignment $m_\mu = (3/2)m_\star$, which lies within 0.61% of the observed mass, therefore has no selecting mechanism in the present functional and is not a prediction.

6.2 The Tau: Hopf-Linked Trefoil Pair Bound by a Muon Quantum

The structural identification rests on a closed muon mechanism

The mass relation $m_\tau = 2m_p - m_\mu$ derived below uses the *measured* muon mass (PDG) as a numerical input, so the predicted 1770.89 MeV is arithmetically correct. But the *structural* identification imports a claim that is **closed as a no-go**: the muon as a $3/2\,m_\star$ core-breathing $\oplus$ Kelvin hybrid eigenmode. Every route to that eigenmode failed — Path A/B, the issue #76 scalar Berry-phase null, the issue #91 spinor lock ($|V| = 0$), and the issue #94 half-quantum-vortex audit (wrong circle + charge quantization). The muon is therefore a *numerical coincidence* with no derived eigenmode (Paper I §The Muon). Consequently the Hopf-link binding-energy reading below has no supporting mechanism: $m_\tau = 2m_p - m_\mu$ should be read as an arithmetic relation among measured masses, and the tau, like the muon, is a numerical coincidence. (It also inherits the proton form-factor calibration F, Paper I §The Proton.) Tracking: issues #76, #91, #93, #94.

The tau sits near $25\frac{1}{2}\,m_\star$ in the empirical α-harmonic clustering ($m_\tau = 25.375\,m_\star$, 0.49%). A former Hopf-link interpretation motivated the arithmetic relation

$$\boxed{m_\tau \;=\; 2\,m_p \;-\; m_\mu \;=\; 2(13.5)m_\star - (3/2)m_\star \;=\; \tfrac{51}{2}m_\star,} \tag{51}$$

which lands on rung $25\frac{1}{2}$ by construction and evaluates, using measured masses, to

$$m_\tau^{\text{pred}} \;=\; 2(938.272) - 105.658 \;=\; 1770.89 \text{ MeV}, \tag{52}$$

to compare with the observed 1776.86 MeV (-0.336%). Using the isospin-averaged baryon mass $(m_p + m_n)/2$ instead of m_p alone gives $m_\tau^{\text{pred}} = 1772.18$ MeV (-0.263%), tightening the agreement since the two trefoils in a Hopf link are not isospin-distinguished in the symmetry channel of the linkage. Either choice is more accurate than the bare ladder coincidence $25\frac{1}{2}\,m_\star = 1785.6$ MeV$/c^2$ ($+0.494\%$).

Topological reading. The (2,3)-trefoil is the simplest non-trivial knot and the topological skeleton of the proton in Paper I [1]. Two trefoils linked through each other once (Hopf link) form the simplest non-trivial two-component link and are the natural next building block above the proton in the SSV defect spectrum. The linkage region carries a shared Kelvin-helicity standing wave; the lowest such mode would have the same eigenfrequency as the muon's core-breathing $\oplus$ Kelvin hybrid — a mode now closed as a no-go (Paper I, §Muon; issues #91, #94) — so the equality of the binding energy with one muon mass is an assumption, not a derived result. The minus sign in (51) is the binding-energy convention: two free trefoils ($2m_p$) bound by a Kelvin quantum lower their total energy by m_μ.

Why this is not a verification path. The relation would require both a stable $(2,3) \oplus (2,3)$ linked solution and a linkage-region eigenmode with binding energy m_μ. Neither exists in the current calculations, and the proposed $m_\varphi = \pm 1$ muon hybrid has already failed. Crossing counts alone do not determine an energy. Consequently no live calculation presently promotes Eq. (51) above coincidence status.

Consequence for the generation problem. The former tower of building-block decompositions (e = toroidal vortex, μ = excited torus, τ = linked trefoils minus a muon quantum) is not a surviving model of the three generations. With the muon eigenmode now a closed no-go (issues #91, #94) the muon and tau identifications remain numerical coincidences, not derived masses. Whether the three lowest topological classes admissible in the SSV functional — pure torus (electron), torus-with-internal-mode (muon, pion), and linked trefoils (tau) — actually realise the charged leptons is open. It sharpens the threefold-generation problem (§6) into a specific topological question: is the SSV functional admissible on the $(2,3) \oplus (2,3)$ link, and at what link spacing does the muon-class binding mode go off-shell?

6.3 The Generation Problem

The Standard Model has three charged-lepton generations (e, μ, τ) paired with three neutrino flavours $(\nu_e, \nu_\mu, \nu_\tau)$. The framework presents a single unified version of this puzzle, since charged leptons and neutrinos in SSV are not independent objects but two facets of the same vortex configuration — the standing toroidal mode and the torsional pulse it emits during reconnection. A first-principles account of the threefold structure should therefore appear simultaneously in:

- **Charged leptons.** The eigenmode spectrum of the $\mathcal{L} + \mathcal{L}_\perp$ functional on the toroidal vortex background should yield the muon eigenfrequency (Paper I open problem 2) and, at the next harmonic, an identification of the tau. Whether the spectrum has exactly three accessible modes below the confinement scale — and why three — is unknown.
- **Neutrinos.** The same eigenmode structure should determine the flavour content of the radiated torsional pulse during reconnection (§5.7, P2 and P3), giving the three neutrino flavours and their oscillation phenomenology as a downstream consequence rather than as a separately tunable sector.

The threefold-generation problem is therefore one problem, not two. Its resolution is deferred to the same 3D minimisation that closes the W-mass gap (§5); the α_G target that this computation also carried is retired — the sub-grain route's monopole charge is formally divergent (#98), and G is at present an empirical input (§7).

7 Gravity: Time-Delay Gradients from Interfering Pressure Waves

Falsification record (issue #119, 2026-06): this section's mechanism is falsified as written; the text is retained as the record

Pre-registered computations (issue #119, Phases 1–2; Paper IV falsification record and result note `issue-119-falsification-and-bath-candidate`) falsify the mutual-radiation Bjerknes mechanism developed below: (i) beyond one wavelength the time-averaged cross-term oscillates in sign with separation — and with $\lambda_p \approx 1.3 \times 10^{-15}$ m every physically realised separation is $\gtrsim 10^{12}$ wavelengths into the radiation zone, so the claimed sign-definite $1/r^2$ force does not exist where gravity is observed (the Step-1 zone-independence claim below was already contradicted by Paper IV's own Appendix-A numerics); (ii) the time-averaged cross-term between breathers of unequal Compton frequency vanishes — no electron–proton gravity, and composite bodies fail the strict $\propto N$ extensivity that Eötvös-class experiments confirm to 10^{-15}; (iii) the static ($\omega = 0$) escape is Yukawa-screened on the healing length. A bath-driven (secondary-Bjerknes) repair is sign-definite and frequency-blind but short-ranged — wrong range for Newton (#119 Phase 2). The Step-3 sub-grain route to α_G is independently retired: the acoustic monopole moment Q_p is formally divergent (#98). **What survives:** the concavity→attraction argument structure; the time-dilation-field observable (phase-blind, hence immune to (i)–(ii)), whose measured source is each defect's *intrinsic* two-term structure — healing-length core plus $\ell^2\rho/2r^2$ circulation halo, with the defect's own Bernoulli density depression as its time-dilation field (#119 H7a/H8). Gravity's coupling G is at present an *empirical input*: the H9 circulation-halo triangle, the last live derivation route, was falsified against the baryonic Tully–Fisher relation (#119 H9). Paper IV carries the surviving architecture.

7.1 The Physical Picture

Gravity is the gradient of a time delay. Every massive defect is a breather (Paper I) — a localised region of Ψ oscillating at its Compton frequency — and it radiates a longitudinal pressure wave into the medium, with amplitude falling as $1/r$ from spherical dilution. A second breather, or a light wave, propagating through that radiated wave interferes with it; the time-averaged interference, through the medium's logarithmic nonlinearity, slows the local rate of change of Ψ. Since physical time is that rate of change (Paper III), the interference imposes a time delay, strongest near the mass and fading with distance. Its gradient bends every trajectory that crosses it toward the region of greatest delay — and that bending is gravity.

A single potential $\Phi(\mathbf{x})$ summarises the delay: the acceleration of free fall is $-\nabla\Phi$, and the fractional slowing of clocks is Φ/c^2. The present section establishes gravity as the fourth force-sector mode and fixes Newton's constant from the medium; the full development of the mechanism — the chain from interference to path bending, and the gravitational consequences that follow — is the subject of Paper IV.

7.2 The Proton as a Pulsating Breather

The source of the radiated wave is the breather itself. In Paper I the proton is a spherical breather mode — a periodically oscillating, localised density perturbation of the superfluid vacuum, sustained by its own knotted vortex skeleton and not by any external driving. Its eigenfrequency is the Compton frequency:

$$f_p = \frac{m_p c^2}{h} \approx 2.27 \times 10^{23}\,\mathrm{Hz}, \tag{53}$$

with corresponding wavelength $\lambda_p = c/f_p \approx 1.3 \times 10^{-15}\,\mathrm{m}$. The breather radiates a longitudinal density wave into the medium at this frequency; the fractional density modulation $x(\mathbf{x}, t) \equiv \delta\rho/\rho_0$ falls as $1/r$ from spherical dilution. Every massive defect does the same, scaled by its mass-energy. Gravity is the interference of these radiated waves, and the next subsection computes it.

7.3 The Interference Cross-Term

The mechanism is contained in the LogSE. Where the radiated waves of several breathers overlap, the local density is $\rho/\rho_0 = 1 + \sum_i x_i$ with $x_i \equiv \delta\rho_i/\rho_0$, and the medium's logarithmic energy term expands as

$$\ln\Bigl(1 + \sum_i x_i\Bigr) = \sum_i x_i - \frac{1}{2}\sum_i x_i^2 - \sum_{i<j} x_i x_j + \cdots \tag{54}$$

The single-wave terms are each breather's own self-energy; the interaction between two breathers A and B is carried by the cross term

$$u_{AB} \;=\; -\,b\,\rho_0\,x_A\,x_B, \tag{55}$$

an *interference* term — the product of two waves. Three properties fix the character of gravity. Its time average $\langle u_{AB}\rangle = -b\rho_0\langle x_A x_B\rangle$ survives only for phase-correlated waves, so coherent breathers gravitate and incoherent disturbances do not. Its sign is fixed negative by the concavity of the logarithm: *gravity is attractive because the medium's nonlinearity is logarithmic.* And it is bilinear — with $x_A \propto m_A/r$ diluted to B's location and $x_B \propto m_B$ — so $\langle u_{AB}\rangle \propto -m_A m_B/r$, the Newtonian potential, with the inverse-square force its gradient. The coefficient is fixed by the breather's acoustic output in §7.5; the full development is given in Paper IV.

7.4 The Bjerknes Mechanism

This time-averaged interaction of pulsating bodies is historically related to the Bjerknes interaction [20]. The 1906 source supports the phase-dependent sign, bilinearity and inverse-square structure for synchronous pulsators in an incompressible treatment; it does not supply the following normalized compressible-medium formula. A later secondary-Bjerknes calculation gives the commonly used volume-rate form

$$\langle F\rangle = -\,\rho_0\,\langle \dot{V}_1\dot{V}_2\rangle\,\frac{1}{4\pi r^2}, \tag{56}$$

with r the separation. The cross-correlation $\langle \dot{V}_1\dot{V}_2\rangle$ is exactly the interference term $\langle x_A x_B\rangle$ of §7.3 written in volume-velocity variables, and the geometric $1/(4\pi r^2)$ is the spherical dilution of a wave in three dimensions — not an independently postulated force law. We use Bjerknes' result and his volume-velocity form below to compute G; "force" is retained only as shorthand for the gradient of the interference. Its reinterpretation — why this is not a force across an inert gap, and why the breathers, as self-sustaining eigenmodes of Ψ, need no external driving — is developed in Paper IV.

7.5 Newton's Constant from the Medium

The quantitative interference calculation uses the breather's acoustic output to fix the coefficient of Φ left open in §7.3. It proceeds in four steps: the wave a breather radiates, the cross-term a second breather feels in it, the sub-grain suppression that makes that coupling weak, and the value of G that results.

Step 1: Acoustic wave of a pulsating breather. The proton pulsates at angular frequency $\omega_p = m_p c^2/\hbar$ with volume oscillation $V(t) = V_0 + \delta V_p \sin(\omega_p t)$, so $\dot{V}(t) = \omega_p \delta V_p \cos(\omega_p t)$. Both the static Bjerknes force and its radiative part are governed by the medium's sound speed c (§3.9). The standard Landau–Lifshitz result for a pulsating sphere [21] gives:

$$\delta P(r,t) = \frac{\rho_0}{4\pi r}\,\ddot{V}\!\left(t-\frac{r}{c}\right) = -\frac{\rho_0\,\omega_p^2\,\delta V_p}{4\pi r}\sin\!\left(\omega_p t - \frac{\omega_p r}{c}\right). \tag{57}$$

Since the proton Compton wavelength $\lambda_p \approx 1.3\times 10^{-15}\,\mathrm{m}$ is smaller than all macroscopic separations, gravitational interactions always occur in the radiation zone ($r \gg \lambda_p$). The original text asserted here that the time-averaged Bjerknes force nevertheless retains its $1/r^2$ form at all distances for in-phase oscillators, "because the static component of the time-averaged force is independent of zone." *That sentence is falsified*: there is no zone-independent static component. The retarded cross-correlation is $\langle x_A x_B\rangle \propto \cos(\omega r/c)/r$, so beyond one wavelength the time-averaged force oscillates in sign with separation — as Paper IV's own Appendix-A simulation shows and the #119 H1a sweep confirmed against the $\cos(2\pi d/\lambda)$ node positions. In the radiation zone, where all real gravitational separations lie, the mechanism is sign-indefinite (falsification record above).

Step 2: The interference cross-term. A second identical proton at separation r is a resonant oscillator at ω_p, driven in phase by the same medium. The time-averaged cross-term of §7.3, in volume-velocity variables, is the average over one cycle:

$$\langle \mathbf{F}_{12}\rangle = -\frac{\rho_0}{4\pi r^2}\langle \dot{V}_1(t)\,\dot{V}_2(t)\rangle\,\hat{r}, \tag{58}$$

with

$$\langle \dot{V}_1\dot{V}_2\rangle = \tfrac{1}{2}\omega_p^2(\delta V_p)^2 > 0, \tag{59}$$

giving an attractive force $\propto 1/r^2$. For two objects of different masses the pulsation amplitude scales as $\delta V \propto m/\rho_0$ (Step 3), so the force is proportional to $m_1 m_2/r^2$ — Newton's law.

Step 3: Pulsation amplitude and sub-grain suppression. The fractional volume excursion of a breather is set by the ratio of its rest-mass energy to the medium's compressive energy over the breather volume:

$$\varepsilon_p \equiv \frac{\delta V_p}{V_0} = \frac{m_p c^2}{\rho_0 c^2 \cdot V_0} = \frac{m_p}{\rho_0 V_0}, \tag{60}$$

where $P_0 \approx \rho_0 c^2$ and $V_0 = \frac{4}{3}\pi \bar{\lambda}_p^3$ with $\bar{\lambda}_p = \hbar/(m_p c)$ the proton reduced Compton wavelength. Since $\delta V \propto m$, the Bjerknes force scales as $m_1 m_2$ as required.

The proton is a *sub-grain* oscillator: its reduced Compton wavelength $\bar{\lambda}_p$ is smaller than the medium healing length $\xi = \hbar/(m_e c)$ by the mass ratio

$$\frac{\bar{\lambda}_p}{\xi} = \frac{m_e}{m_p} \approx \frac{1}{1836.15}. \tag{61}$$

The medium cannot resolve structure below ξ; consequently the proton couples to the long-range density wave with a suppressed *acoustic monopole moment* Q_p. For a sub-grain spherical oscillator, coupling to wavelengths $\lambda \gg \xi$ is suppressed by the ratio of source volume to grain volume:

$$Q_p \sim \delta V_p \cdot \left(\frac{\bar{\lambda}_p}{\xi}\right)^3 = \delta V_p \cdot \left(\frac{m_e}{m_p}\right)^3. \tag{62}$$

The coupling hierarchy across the three stable objects is:

$$\eta_\gamma = 1 \;\gg\; \eta_e \sim \alpha \;\gg\; \eta_p \sim \left(\frac{m_e}{m_p}\right)^3, \tag{63}$$

where the photon couples with unit efficiency (it *is* the medium wave), the electron couples through its toroidal topology (suppressed by α), and the proton couples as a sub-grain monopole (suppressed by $(m_e/m_p)^3$). The original conclusion — that gravity is weak because the proton is small relative to the medium grain — *does not survive*: the suppression factor belonged to the falsified mechanism, and the monopole moment Q_p it required was subsequently shown to be formally divergent, with no box-stable value (#98). The weakness of gravity — $\alpha_G \sim 10^{-39}$ — is at present the sector's unsolved hierarchy problem, not a derived consequence.

Step 4: Newton's constant from the medium. Substituting Q_p into (58) and matching to $\langle F_{12}\rangle = Gm_p^2/r^2$:

$$G = \frac{\rho_0\, \omega_p^2\, Q_p^2}{8\pi\, m_p^2}. \tag{64}$$

A full evaluation of Q_p requires the 3D proton breather profile and is deferred to a companion paper. A structural consistency check uses the gravitational coupling constant $\alpha_G \equiv Gm_p^2/(\hbar c)$ and the Paper I proton mass, $m_p = N_p\, m_e/\alpha$ with $N_p = N_Y F \approx 13.44$:

$$G = \frac{\alpha_G\, \hbar c\, \alpha^2}{N_p^2\, m_e^2}. \tag{65}$$

This is a structural consistency check: α_G is taken from CODATA rather than derived from first principles, confirming that the Paper I proton mass is internally coherent with the gravitational sector. Inserting $\alpha_G = 5.906 \times 10^{-39}$ (CODATA) and $N_p = 13.44$:

$$G_{\text{pred}} = \frac{5.906 \times 10^{-39}\; \hbar c\, \alpha^2}{(13.44)^2\, m_e^2} = 6.634 \times 10^{-11}\, \text{m}^3\, \text{kg}^{-1}\, \text{s}^{-2}, \tag{66}$$

within 0.6% of $G_{\text{obs}} = 6.674 \times 10^{-11}\, \text{m}^3\, \text{kg}^{-1}\, \text{s}^{-2}$. The residual is consistent with the imprecision in N_p (form factor $F \approx 4.47$ is numerically determined; see Paper I).

Open problem: G has no live derivation route (status after #119/#98)

$\alpha_G = 5.9 \times 10^{-39}$ is at present an *empirical input*. Equation (65) expresses G in terms of $\{\hbar, c, \alpha, m_e, \alpha_G\}$ and matches CODATA to 0.6% when α_G is supplied; that is a numerical consistency check of the Paper I mass ladder, not a derivation. Every route that was to supply α_G from the medium is now closed: (i) the sub-grain suppression factor $(\bar{\lambda}_p/\xi)^n$ belonged to the mutual-radiation mechanism falsified above, and its monopole charge Q_p is formally divergent — the LogSE healing tail decays as $1/r^2$, so $\int r^2\,\text{deficit}\,dr$ grows with box size and no box-stable value exists (#98, closed resolved-negative); (ii) the circulation-halo triangle $v_{\text{flat}}^2 = G\rho_0\Gamma^2/(2\pi c^2)$, solved against the baryonic Tully–Fisher relation on the natural baryon-locked chain, misses α_G by $\sim$35 orders of magnitude and predicts BTFR slope 3/2 instead of 4 (#119 H9, falsified on both pre-registered rules); (iii) naive Sakharov scaling at the SSV grain overshoots by $\sim$39 orders in the opposite direction. The 0.6% figure also inherits the SSV-I F-cutoff calibration of N_p (issue #66): $N_p = 13.44$ is the midpoint of the two-fine-grid bracket 13.28–13.62, and since $G \propto N_p^{-2}$ the inherited spread on G_{pred} is -3.2% to $+1.8\%$, five times the quoted residual. The weakness of gravity is the sector's hierarchy problem, fully exposed.

Candidate common carrier at the sound speed. The phase-channel disturbance propagates at c, so it is the remaining candidate carrier after the chiral-shear mode is eliminated. The current functional has not derived electromagnetic transverse polarizations, Maxwell source matching, or a gravitational tensor mode from this scalar branch. It therefore does not yet establish that electromagnetic and gravitational radiation share a carrier merely because both observed speeds are compatible with c.

This is consistent with the GW170817 observation. The neutron-star merger GW170817 produced gravitational and electromagnetic signals separated by $\sim 1.7\,\text{s}$ [22]. That timing tightly constrains the fractional propagation-speed difference only after allowing for an uncertain intrinsic emission lag. Source processes such as jet launch, cocoon formation and breakout can contribute to the lag, but the cited observation does not determine that decomposition or show that the delay is entirely source physics. The chiral-shear sector mediates the static near-field Coulomb interaction between vortices and supports no propagating linear mode at all (#138; §3.9); it is not the carrier of any astrophysical radiation.

The spatial sector after #138: the magnon is not the missing carrier either. The #129 computation (Paper IV) established that the bare phase channel delivers gravity's time sector and none of its spatial sector ($\gamma_{\text{eff}} = 0$): a transactional carrier — one whose transport consults the local medium state hop by hop, paying the bilateral A^2 availability tax — was the surviving candidate mechanism, and the CP^1 internal-direction channel was the candidate carrier. The #138 battery answered this (its Q3) in the *negative*, and the negative is exact: over any static equilibrium background, the magnon's effective potential $V + b\ln\rho_b = \mu$ is a global constant — the same symmetry-protected cancellation that produced the #129 null (measured $\gamma_{\text{magnon}} \approx 10^{-4}$ with the instrument validated on an explicit spin-channel potential). The $\hat{n}$ channel is exactly as non-transactional as sound. As of #138, the present medium therefore contains *no candidate carrier for the spatial sector of gravity*: the A^2 doubling demanded by lensing data is mandatory structure that neither the phase channel, nor the chiral-shear sector, nor the CP^1 channel supplies. This negative is recorded at full strength.

7.6 Scope: the Full Treatment in Paper IV

The present section proposed gravity as the fourth force-sector mode — the interference of breather pressure waves, building a time-delay gradient that bends trajectories — and that proposal is falsified as written, with the section retained as the record (falsbox above). What hands off to Paper IV is the surviving architecture: the time-dilation field as the observable and the intrinsic defect source. The conceptual development is carried out in Paper IV: the formal account of update capacity, the four gravitational consequences (free fall, time dilation, light deflection, redshift), the universality of the coupling, the absence of a graviton, the bridge to emergent geometry (Paper VII-b), and the connection to the recording medium of Paper III.

8 Unified Picture: One Medium, Four Modes

Table 2: The four forces and the Higgs mode as facets of the saturated superfluid vacuum. The four forces describe how defects in the medium interact; the Higgs row is structurally different — it is the medium itself oscillating, not an interaction between defects.

Sector	**Mode type**	**Source**	**Carrier**	**Ran**
Electromagnetism	Transverse flow	Chirality ($\pm$)	Transverse phonon	∞
Strong	Surface tension	Topological charge	Phase oscillation	$\sim a/$
Weak	Reconnection	Core overlap	Amplitude mode (W, Z)	$\sim \xi$
Gravity	Time-dilation gradient†	Mass-energy (intrinsic defect loading)	Longitudinal density field	∞
Higgs	Amplitude mode	The medium itself	$\delta\rho$ oscillation	$\sim \xi_{\mathrm{E}}$

We can now survey the four forces from a single vantage point. Table 2 summarises the correspondence and adds the Higgs as the one remaining mode of the medium that is not an interaction between defects. (†The gravity row reflects the post-#119 status: the original entry, pressure-wave *interference*, is the falsified mutual-radiation mechanism of §7; the surviving reading is the time-dilation gradient sourced by each defect's intrinsic loading of the medium, developed in Paper IV.)

Several structural features of this table deserve comment.

Range. The infinite-range forces (EM and gravity) are both mediated by massless phonon modes — transverse and longitudinal, respectively. The short-range forces involve massive excitations: the gluon acquires an effective mass from the string tension at distances exceeding a/α, and the W/Z are the high-energy amplitude modes of the core. The hierarchy of ranges $r_W \ll r_{\text{strong}} \ll r_{\text{EM}}, r_{\text{gravity}}$ follows directly from the hierarchy $\xi \ll a/\alpha \ll \infty$.

Coupling strengths. Three of the four couplings are fixed by the same two numbers, α and $m_\star = m_e/\alpha$: the strong coupling is ~ 1 (order-unity surface tension, no small factor), the electromagnetic coupling is $\alpha \approx 1/137$ (transverse-to-longitudinal wave speed ratio), and the weak coupling is α^2 (two powers of the core penetration suppression). The gravitational coupling $Gm_p^2/\hbar c \approx 6 \times 10^{-39}$ is the exception: it is an empirical input with no live derivation route (gapbox, §7.5) — the framework's hierarchy problem.

Universality of gravity. Every other force is selective: EM acts on chirality, strong on topological charge, weak on core overlap. Gravity couples to the one property all defects share — they load the medium with mass-energy, and every process advances by updating it. The response side of this universality (everything falls alike in a time-dilation gradient) is derived in Paper IV; that the *source* side scales strictly as total mass-energy for composite bodies is an open requirement on the intrinsic-source derivation (#119).

Defects vs. canvas. The four force rows of Table 2 share a common pattern: each describes how some property of a defect (chirality, topological charge, core overlap, mass-energy) sources a pressure response of the medium that another defect feels. The Higgs row breaks this pattern. Its source is not a defect but the medium itself; its carrier is the same density mode that, when sourced by a moving breather, transmits gravity — but intrinsically rather than as an interaction. A reader who finds the Higgs "out of place" in a force-unification table is correctly identifying the asymmetry: the Higgs is not a fifth force, it is the medium's own amplitude mode made briefly visible. Its inclusion makes explicit that the same Ψ-field substrate produces both the four-force interaction structure and the matter / canvas distinction that underlies it.

Particle content on the same substrate. The same medium produces the full matter content of the Standard Model: electrons as toroidal vortex rings, baryons as knotted vortex breathers, charged leptons *conjecturally* as higher breathing modes (§6; unconfirmed — the muon and tau masses are numerical coincidences), neutrinos as torsional radiation from reconnection (§5.7), and the discrete symmetry structure (P-violation, the absence of exact CP, the absence of a strong CP problem) as direct consequences of the medium's chirality and recorded history (§5.6). Particles, forces, and symmetries are not three separate sectors with mutual couplings; they are three different ways of asking questions about the same hydrodynamic substrate.

No new constants. The framework introduces no coupling constants beyond those already fixed by the mass spectrum of Paper I. The four forces, the Higgs sector, the particle content, and the discrete-symmetry structure share the same parameters $\{c, \alpha, m_\star\}$; they are not independent sectors with mutual couplings, but four different windows into the same hydrodynamic pressure field.

9 Testable Predictions

The framework makes a number of predictions that go beyond what is currently explained within the Standard Model. We organise them by sector to make clear which experiments would test which structural claims.

Force-sector and gravitational predictions

1. **Gravitational-wave polarisation — falsified as written (#129 GW-POL, 2026-06).** The longitudinal acoustic phonon of the superfluid is spin-0, and there is no graviton in SSV. The earlier expectation — that gravitational radiation would nonetheless carry an *effective* spin-2 polarisation from a transverse-traceless projector in the linearised acoustic metric — is refuted by direct computation (`gw_polarization`). The acoustic (Unruh) metric is conformal in its spatial part plus a vector flow term and has *no* transverse-traceless component, so the wave is purely helicity-0 (breathing + longitudinal): its ψ-harmonic overlap with the tensor $+/\times$ pattern is exactly zero, placing SSV in the polarisation class that LVK *disfavours* (GW170814, GW170817). Worse for detectability, the isotropic spatial metric — the same structure that lenses light correctly ($\gamma = 1$) — cancels in the differential interferometer readout ($F_{\text{breathing}} = -F_{\text{longitudinal}}$); the residual is an $\mathcal{O}(fL/c)$ longitudinal leak ($\approx 2.9 \times 10^{-3}$ at 100 Hz, 4 km), so a true SSV wave read under tensor templates is inferred $\sim$340$\times$ too distant (GW170817's 40 Mpc $\to \approx$14 Gpc). The one clean pass: SSV emission carries no monopole or dipole (mass/momentum conservation, Lighthill), so it survives the Hulse–Taylor $\sim$0.1% dipole bound, and $c_{\text{gw}} = c$ holds natively (GW170817). The negative is contingent on the present U(1)-scalar order parameter: a richer (tensorial) order parameter, or the Weinberg–Witten preferred-frame loophole opened by the

substrate "now", could in principle supply a transverse-traceless mode — the surviving-version requirement, recorded in `papers/SSV-IV/results/gw-pol-issue129.md`.

2. **Gravitational Casimir correction.** The Bjerknes force between a proton and any other mass includes a next-to-leading-order correction from the *quantum pressure* term, analogous to the Casimir effect. At sub-micron separations, this produces a deviation from the $1/r^2$ law of order $(\delta r/r)^2 \sim (a/r)^2$, where $a \sim 10^{-15}$ m is the vortex core size. This is below current precision for laboratory experiments but could in principle be detected at hadron-scale separations in neutron scattering experiments.

3. **Gravitational-wave memory.** The gravitational-wave memory effect (a permanent shift in detector test-mass separation following the passage of a strong GW signal) is predicted to be present at the Christodoulou level — not as a peculiar property of asymptotic spacetimes but as a direct consequence of a medium that records (Paper IV). The framework sets the same magnitude as conventional GR; the structural claim is that memory is the natural state of affairs in SSV, not an exotic signal. Distinguishing SSV from GR on this point requires precision detection of the fluctuation spectrum around the memory mean, which is beyond current sensitivity.

4. ***W*-mass scale derived; $\mathcal{O}(1)$ prefactor open.** The reconnection cap inherits the electron ring scale $R^* = \xi/\alpha$ (Paper I), so the cap energy $m_W \sim P_0 R_{\rm cap}^2 \xi \sim m_e/\alpha^2$ has a *derived* $1/\alpha^2$ scale (≈ 9.6 GeV). The Rayleigh-jet cap radius $R_{\rm cap} = \phi R^*$ (§5.4, equation (41)) then gives $m_W \approx 78.9$ GeV — within 1.8% — with the prefactor $\pi\phi^2$ an $\mathcal{O}(1)$ coincidence (ϕ matches the cap aspect to 0.9%). An earlier attempt to derive $R_{\rm cap}$ from a local chiral-shear bending equilibrium (§5.4, equation (44)) is retired: a local $\mathcal{L}_\perp$ curvature correction is structurally α^{-2} and undershoots the ring scale by exactly one power of $1/\alpha$ (the "factor 232"), confirming that the cap scale is set by ring inheritance, not bending (issue #105).

5. **Running of α at high energy.** The chiral-shear coupling α is a medium property, not a bare constant. At momentum scales approaching $E_\star = m_e c^2/\alpha$, the chiral-shear sector begins to couple to the longitudinal modes through the nonlinear term in the Gross–Pitaevskii equation. This produces a logarithmic running of α that should match the QED renormalisation group equation — a self-consistency check that has not yet been completed in detail.

Particle-content predictions

6. **Higgs as condensed-matter analogue.** Since the Higgs is identified as the amplitude mode of the same order parameter that carries every other particle (§5.3), its phenomenology should match that of the amplitude mode in cold-atom condensates near saturation [8], scaled up by the medium parameters. The decay channels of the Higgs reflect the eigenmode structure of the medium's amplitude oscillations; small deviations from Standard Model branching ratios at high precision (above current LHC sensitivity) would be expected if the amplitude mode is genuinely a hydrodynamic excitation rather than a fundamental scalar. The framework does not predict a specific deviation magnitude; the structural claim is that the Higgs is not a separately tunable sector.

7. **No fourth-generation lepton.** The framework conjectures that the toroidal vortex background supports a finite eigenmode spectrum; the muon and tau would be placeholders for the lowest two ring-breathing modes (§6) — though this eigenmode identification is unconfirmed (issues #91, #94) and the masses are coincidences. Whether the spectrum has exactly three accessible modes below the confinement scale is open, but the threefold Standard Model generation structure — if it follows from this eigenmode structure —

would imply that searches for a fourth charged lepton or a fourth neutrino flavour should return null at all accessible energies. Any positive result would falsify the eigenmode interpretation of the lepton generations.

8. **Neutrino emission only in topology-changing events.** In the SSV picture, neutrinos are torsional pulses emitted only when vortex topology is rewritten — i.e. in weak-sector reconnections. No neutrino emission accompanies any purely electromagnetic or strong-sector process, including high-energy QED vertices, hadronic resonance decays that proceed through the strong channel, or coherent photon scattering, regardless of available phase space. This is a sharper statement than the Standard Model selection rule (which is a flavour-current statement); here it is a structural impossibility, since no torsional pulse can be sourced without a reconnection. Any observation of neutrino emission unaccompanied by topology change in the source would falsify the framework.

9. **No magnetic monopoles.** The magnetic field is defined as $\mathbf{B} \equiv c_\perp^{-1}\nabla\times\mathbf{v}_\perp$ (Section 3.6), so $\nabla\cdot\mathbf{B} \equiv 0$ is a vector identity, not a postulate. A magnetic monopole would require a point-like topological defect sourcing the transverse velocity as a scalar potential — but the GPE supports only vortex-line defects (which source the *rotational* part of $\mathbf{v}_\perp$, carrying electric charge). The absence of magnetic monopoles is a topological theorem of the framework: no object with the required topology can exist in a superfluid order parameter in three spatial dimensions. All current and future monopole searches should return null results.

Discrete-symmetry and CP-sector predictions

10. **Neutron electric dipole moment from reconnection memory.** The strong CP problem dissolves in SSV (§5.6): there is no CP-symmetric reference vacuum, so $\theta_{\rm QCD}$ is not a free parameter to be tuned. The reconnection-memory derivation of §5.6 gives the leading-order structural form

$$d_n \;=\; \kappa_\chi\, e\, R_n\, \frac{n_B}{n_\gamma} \;\approx\; \kappa_\chi \;\times\; 5\times10^{-23}\, e\!\cdot\!\mathrm{cm},$$

with κ_χ a dimensionless chiral-shear coupling deferred to the converged 3D reconnection-barrier minimisation (issue #30). At the natural scale $\kappa_\chi \sim \alpha^2/\pi$ this gives $d_n \sim 10^{-27}\, e\cdot$cm, one decade below the present bound $|d_n| < 1.8\times10^{-26}\, e\cdot$cm [9] and within reach of nEDM/n2EDM at $\sim 10^{-28}\, e\cdot$cm. The same chirality memory density χ underlies both this prediction and the cosmological baryon asymmetry, tying two distinct CP observables to a single framework coupling.

11. **No axion required.** The Peccei-Quinn mechanism and its associated axion are introduced in the Standard Model to explain the smallness of $\theta_{\rm QCD}$. Since SSV does not have a $\theta_{\rm QCD}$ parameter to begin with, no axion is required, predicted, or expected. Detection of an axion at any mass would not falsify SSV directly — axions could exist as additional medium excitations not yet identified in the framework — but their detection would remove the SSV's most distinctive structural prediction in the strong-CP sector.

12. **Baryon asymmetry from chiral seeding.**

    The observed baryon-to-photon ratio $n_B/n_\gamma \sim 6\times10^{-10}$ is interpreted as the surviving recorded chirality of the seeding event (§5.6). Quantitatively, the prediction is that this ratio should follow from the same chirality bias that determines the CKM phase — i.e. from a single number measured in two completely different experiments. This is a sharp consistency prediction: whatever calculation eventually produces the CKM phase from reconnection geometry should also produce the baryon-to-photon ratio. No separate cosmological CP-violating sector should be required.

Falsification structure

The above predictions are not equally falsifiable, and it is worth being explicit. The structural predictions (no monopoles, no axion required, neutrinos only from topology change, no fourth-generation leptons) admit clean falsification: a single confirmed positive observation in any of these categories would invalidate the framework as currently formulated. The quantitative predictions (gravitational-wave polarisation corrections, W mass, Higgs phenomenology, n_B/n_γ) require performing the 3D $\mathcal{L} + \mathcal{L}_\perp$ minimisation before they can be compared against observation; their failure mode is not a single-experiment refutation but the inability to reproduce known numbers from the geometry. The gravitational-wave memory and gravitational Casimir predictions are quantitatively close to GR/Standard-Model expectations and would require precision improvements beyond current capability to discriminate.

10 Bridge to Thermodynamics and Irreversible Time

The same superfluid order parameter Ψ that underlies the four forces also governs irreversible thermodynamics. One further connection to a subsequent paper is worth flagging here, beyond the forward-pointer already given in §5.6 (CP and memory).

Time and entropy (Paper III). The two flagged places where the present paper meets the irreversibility framework — the symmetry/process distinction in CP, and the time-averaging step in Bjerknes gravity — are not isolated remarks but the visible edges of a single substrate question: in a medium whose state at any moment is the record of all reconnections that preceded it, what counts as an instantaneous quantity? Paper III treats this question directly: the arrow of time is identified with the growth of the topological tangle, and the second law is the statement that the medium accumulates record monotonically. The cosmological side of the same picture — the CMB thermal spectrum read as acoustic noise integrated from all reconnection events since the seeding — is the subject of a separate future paper (Paper IX, on the primordial phonon bath), not of the present paper. The force-sector mean-field treatment of the present paper is the static envelope of that fuller dynamics.

11 Discussion and Open Problems

The framework presented here is mechanically complete for the force sector: each of the four interactions has a definite hydrodynamic origin, and the extensions added in this paper — the Higgs as the canvas's amplitude mode, neutrinos as torsional radiation, charged leptons as conjectured higher breather modes (unconfirmed), and CP violation as a fluctuation residue in a medium that records — fill out the particle content and the discrete-symmetry structure of the Standard Model on the same underlying hydrodynamic substrate. No additional fields, gauge groups, or coupling constants are introduced beyond those fixed by the mass spectrum of Paper I.

What remains is computation, not conceptual machinery. The open problems collected here divide into two groups: a single core calculation on which most of them depend, and a small number of structural questions that are genuinely independent.

The core open computation

The most significant feature of the present framework's open-problem landscape is that it collapses several apparently independent puzzles of the Standard Model into a single deferred computation: the full 3D time-dependent minimisation of the $\mathcal{L} + \mathcal{L}_\perp$ functional during a vortex reconnection event. The following observables are all determined by the geometry of that minimisation, and none can be computed in isolation from the others:

1. **The W boson mass.** The scalar estimate $E_{\text{recon}} \sim m_e/\alpha^2 \approx 9.6\,\text{GeV}$ accounts only for the quantum-pressure barrier that initiates reconnection. The dominant cost is $E_{\text{caps}} \sim P_0 \cdot \pi R_{\text{cap}}^2 \xi$ — holding open the two fully-suppressed end-caps against the saturation pressure P_0. The formula $R_{\text{cap}} = \phi\,\xi/\alpha$ (equation (41)) gives $m_W \approx \pi\phi^2 m_e/\alpha^2 \approx 78.9\,\text{GeV}$ — within 1.8% of the observed value. The ring scale motivates $m_W \sim m_e/\alpha^2$, but the $\mathcal{O}(1)$ prefactor is not predicted: ϕ was selected after back-solving the observed W mass and is a 0.9% numerical coincidence. The local $\mathcal{L}_\perp$ bending equilibrium that would have set R_{cap} is retired (structurally α^{-2}, the "232×" undershoot is one power of $1/\alpha$).

2. **The neutrino sector (P1–P5 of §5.7).** Mass, three-flavour structure, oscillation phenomenology, Dirac/Majorana character, and the Fermi constant G_F all follow from the same reconnection geometry: the torsional pulses radiated during the event carry whatever flavour structure, mass spectrum, and mixing the linearised GP+chiral operator supports as eigenmodes around the 3D reconnection background.

3. **The charged-lepton generations (§6).** The muon eigenfrequency (Paper I open problem 2) and the tau identification (currently a bare ladder coincidence) are eigenmodes of the same operator, evaluated on the toroidal vortex background rather than the reconnection background. Paper I and Paper II treat them separately for expository reasons; they are mathematically the same problem.

4. **CP violation observables (§5.6).** The CKM phase, the kaon ε-parameter, the B-system CP asymmetries, the neutron electric dipole moment, and the baryon-to-photon ratio $n_B/n_\gamma \sim 6 \times 10^{-10}$ are all consequences of the chirality bias in reconnection outcomes. In a medium that records, these are not small departures from a CP-symmetric reference dynamics; they are the visible component of asymmetries present in every event, made measurable when reconnection geometry happens to amplify them past detection threshold.

5. **The strong sector at two loops.** Identification of colour charge with phase balance at Y-junctions, and the derivation of the QCD β-function from quantum-pressure softening, should be carried to two-loop order using the same 3D reconnection treatment. Comparison with precision QCD measurements provides an independent calibration of the reconnection geometry.

These are not five independent open problems but five facets of the same computation. Each of the corresponding Standard-Model parameters (the electroweak scale, the PMNS matrix, the lepton mass ratios, the CKM phase, Λ_{QCD}) is currently fitted to data; the framework's claim is that all are determined by the geometry of one event, computable in principle without empirical input. This is a sharp, falsifiable claim — the geometry will produce specific numbers that either match observation or do not.

Structurally separate open problems

Three further open problems stand outside the core computation, in the sense that their resolution requires different tools.

6. **Newton's constant: α_G from the proton breather.** The structural formula (65) reproduces G_{obs} to 0.6% using the Paper I proton mass. The remaining task is computing the sub-grain acoustic monopole suppression factor $(\bar{\lambda}_p/\xi)^n$ from the 3D *static* proton breather profile, which would yield $\alpha_G = Gm_p^2/(\hbar c) \approx 5.906 \times 10^{-39}$ without empirical input. This is a different computation from the reconnection one (a static eigenmode rather than a dynamical event), but uses the same $\mathcal{L} + \mathcal{L}_\perp$ functional.

7. **Spin-2 polarisation from linearised Madelung.** A systematic post-Newtonian expansion of the Madelung equations around a pulsating source is needed to confirm the spin-2 angular pattern of gravitational radiation from the longitudinal acoustic carrier. This is a perturbative calculation around the static gravity picture of §7, independent of the reconnection geometry.

8. **The canvas-grain hierarchy (§5.3).** The Higgs mass corresponds to a canvas grain length $\xi_{\mathrm{EW}} \approx 1.6 \times 10^{-18}$ m, roughly 10^5 times finer than the vortex healing length ξ set by m_e. The framework does not solve the hierarchy problem; it restates it as the geometric question of why the medium has two natural length scales separated by this factor. Note that the Standard Model fine-tuning instability does *not* transfer: the cutoff ξ is itself a property of the medium, so there are no quadratically divergent radiative corrections that would drag m_H to a higher scale. The problem is structural, not a fine-tuning.

Boundary with Paper III

Two specific places in this paper depend on operations whose finer-resolution treatment is the subject of Paper III on irreversible thermodynamics. They are flagged here for completeness; neither obstructs the mean-field claims of the present paper.

9. **Time-averaging in a recording medium (Paper IV).** The Bjerknes time-averaging that produces Newton's law is a classical operation; in a medium that accumulates a record of every reconnection event, the corrections to the leading-order static result carry information about pulsation history, and reproduce the gravitational-wave memory effect (Christodoulou) as a natural consequence.

10. **Discrete symmetries at the level of physical processes (§5.6).** The distinction between formal symmetries of the $\mathcal{L} + \mathcal{L}_\perp$ functional and symmetries of physical processes — which are broken by the medium's accumulated record — is the conceptual root of irreversibility. Paper III makes this distinction the foundation of its treatment of time and entropy.

What is closed

It is worth stating explicitly what the present paper closes, since the length of the open-problems list could otherwise mislead. The mean-field mechanical content of all four forces is fixed: each is a calculable pressure response of the saturated medium, with coupling constants determined by $\{c, \alpha, m_\star\}$ from Paper I. The Higgs is identified with the amplitude mode of the same order parameter Ψ, with no separately postulated field. Magnetic monopoles are topologically forbidden, not merely absent. The ultraviolet divergence of perturbative quantum gravity does not arise. The strong CP problem and the cosmological constant fine-tuning problem do not arise. Parity violation, the $(V - A)$ structure of the weak current, the equivalence principle, the universality of gravitational coupling, and the impossibility of magnetic monopoles all follow from the single hydrodynamic substrate without additional assumption.

The remaining open problems are computational, not foundational. Each has a definite formulation within the framework already established here and in Paper I, and a clear path to numerical resolution. The framework's most distinctive structural feature is the unification of what the Standard Model treats as independent puzzles — the W mass, the neutrino sector, the charged-lepton generations, and the CP-violation observables — into facets of a single 3D minimisation. Whether that minimisation, when finally performed, produces the right numbers is the empirical test the framework sets for itself.

12 Conclusion

We have proposed that the four fundamental forces emerge from a single superfluid vacuum as distinct mechanical modes. The present paper is a programmatic and structural report; the precise epistemic status of each sector is summarised below.

What the paper establishes. The mechanical origin of the electromagnetic near-field interaction — chiral flow gradients around toroidal vortices, with Coulomb's law and the Biot–Savart force following from the Bernoulli pressure field — is reached at the level of structural mechanism using the chiral-shear coupling α and the medium parameters fixed in Paper I. The photon-speed question that long shadowed this sector is resolved (#138; §3.9): the chiral-shear sector has no propagating linear mode. The Goldstone phase mode at c is the remaining carrier candidate, while the two-polarization Maxwell derivation is still an open obligation. The earlier three-strand quark/colour interpretation is retired because the trefoil is one closed curve and does not furnish $SU(3)$ colour. A quantitative strong-force and confinement mechanism remains open.

What the paper derives, and what stays coincidence. For the weak force, the reconnection cap inherits the electron ring scale $R^* = \xi/\alpha$ (Paper I), so the W mass scale $m_W \sim m_e/\alpha^2$ is *derived*; the Rayleigh-jet cap radius $R_{\text{cap}} = \phi R^*$ then recovers m_W to 1.8%, with ϕ an $\mathcal{O}(1)$ cap-aspect coincidence (0.9%). The earlier local-chiral-shear bending-equilibrium route to R_{cap} is retired: a local $\mathcal{L}_\perp$ curvature correction is structurally α^{-2} and undershoots the ring scale by one power of $1/\alpha$ (the "factor 232"), which confirms that the cap scale is set by ring inheritance rather than bending.

What the gravity falsification leaves. The mutual-radiation Bjerknes route does not give viable gravity: its far-zone force changes sign, unequal-frequency sources decouple, and composite-body extensivity fails (#119). The relation $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ uses α_G from CODATA and matches the Paper I proton mass to 0.6%. This is an algebraic consistency check, not a derivation of G; no live first-principles route to G remains in this paper. Gravitational and electromagnetic radiation share the Goldstone (phase) channel of the medium and therefore propagate at the medium's sound speed c, consistent with GW170817; the chiral-shear sector supports no propagating linear mode and is the static near-field stiffness (#138; §3.9).

What the paper proposes as candidate geometry. The Higgs is recast as two distinct features: a structural amplitude mode of the LogSE Lagrangian (gapped at $m_H c^2$, present everywhere) and a derived denting event with energy $E_H = P_0 V^*$. Charged leptons were proposed as higher breathing modes of the toroidal electron vortex; this dynamical identification is unconfirmed (issues #91, #94) and the muon and tau masses are numerical coincidences. Neutrinos are presented as a candidate torsional-radiation geometry; this candidate currently contradicts the established phenomenology in five ways (§5.7), and the section indicates where in the medium a neutrino-like object might live rather than offering a competing phenomenology.

What the paper raises as a programmatic question. The CP discussion observes that, in a medium that accumulates a record of every reconnection, no discrete symmetry of the dynamics need survive at the process level. This is a conceptual reframing. Whether it can be made quantitatively compatible with the observed precision of CP-even processes (small θ_{QCD}, small neutron EDM, predominantly CP-even behaviour across most kaon and B-meson observables) requires a suppression mechanism that is currently open.

What the paper identifies as the genuine open problems. The genuine hierarchy question in SSV — why the toroidal vortex ring has equilibrium radius $R^* \approx 137\,\xi$ while the knotted breather's core is $\bar{\lambda}_p \approx \xi/1836.15$, both solutions of the same field equations — is stated but not solved. Several apparently distinct open problems of the Standard Model — the W mass, the neutrino sector, the charged-lepton generations, and the CP observables — collapse, in this framework, into facets of a single 3D minimisation of the $\mathcal{L} + \mathcal{L}_\perp$ functional during a vortex reconnection event. The defect-spectrum hierarchy is an additional target of that minimisation. Whether the minimisation, when finally performed, produces the right numbers is the empirical test the framework sets for itself. That computation is the principal task left to subsequent work.

Scope. All claims above use the medium parameters first fixed by the mass spectrum of Paper I: the sound speed c, the fine-structure constant α, and the base mass scale $m_\star = m_e/\alpha$. No additional fields, symmetries, or coupling constants are introduced in this paper. The framework is presented as a continuum effective theory valid down to $\ell_{\text{grain}} \equiv \bar{\lambda}_p \approx 2.1 \times 10^{-16}\,\text{m}$, the scale of the heaviest stable defect. What lies below this scale is a separate question, in the same sense that Navier–Stokes makes no claim about sub-molecular structure. The mean-field force-sector treatment of the present paper is the static envelope of a fuller picture in which the medium accumulates a record of every reconnection event. Paper III treats that record directly, identifying irreversible time and entropy with the growth of topological complexity.

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A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (`https://github.com/StigNorland/SVT`): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #30 — `https://github.com/StigNorland/SVT/issues/30`
- #66 — `https://github.com/StigNorland/SVT/issues/66`
- #76 — `https://github.com/StigNorland/SVT/issues/76`
- #81 — `https://github.com/StigNorland/SVT/issues/81`

- #83 — https://github.com/StigNorland/SVT/issues/83
- #87 — https://github.com/StigNorland/SVT/issues/87
- #91 — https://github.com/StigNorland/SVT/issues/91
- #92 — https://github.com/StigNorland/SVT/issues/92
- #93 — https://github.com/StigNorland/SVT/issues/93
- #94 — https://github.com/StigNorland/SVT/issues/94
- #97 — https://github.com/StigNorland/SVT/issues/97
- #98 — https://github.com/StigNorland/SVT/issues/98
- #105 — https://github.com/StigNorland/SVT/issues/105
- #119 — https://github.com/StigNorland/SVT/issues/119
- #129 — https://github.com/StigNorland/SVT/issues/129
- #138 — https://github.com/StigNorland/SVT/issues/138
- #180 — https://github.com/StigNorland/SVT/issues/180
- #183 — https://github.com/StigNorland/SVT/issues/183
- #184 — https://github.com/StigNorland/SVT/issues/184
- #196 — https://github.com/StigNorland/SVT/issues/196
- #198 — https://github.com/StigNorland/SVT/issues/198

Code (each pinned to the commit that last modified it).

- instruments/paper_ii/chiral_dispersion.py @ 96b16ea — https://github.com/StigNorland/SVT/blob/96b16ea/instruments/paper_ii/chiral_dispersion.py
- instruments/paper_ii/lperp_core_integral.py @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/instruments/paper_ii/lperp_core_integral.py
- instruments/paper_ii/ssv_ii_ab_audit_2026.py @ f4506d2 — https://github.com/StigNorland/SVT/blob/f4506d2/instruments/paper_ii/ssv_ii_ab_audit_2026.py
- instruments/paper_ii/vortex_cap_mass.py @ 664a338 — https://github.com/StigNorland/SVT/blob/664a338/instruments/paper_ii/vortex_cap_mass.py
- instruments/paper_ii/wmass_cap_scale_resolution.py @ 02b6001 — https://github.com/StigNorland/SVT/blob/02b6001/instruments/paper_ii/wmass_cap_scale_resolution.py
- instruments/paper_iv/gw_polarization.py @ cda2581 — https://github.com/StigNorland/SVT/blob/cda2581/instruments/paper_iv/gw_polarization.py

Reports (result notes, pinned to their last commit).

- papers/SSV-I/results/solver/static-3d-closure-status-2026-05-28.md @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/papers/SSV-I/results/solver/static-3d-closure-s md
- papers/SSV-II/results/chiral-dispersion-issue138.md @ dddc3b8 — https://github.com/StigNorland/SVT/blob/dddc3b8/papers/SSV-II/results/chiral-dispersion-issue13 md
- papers/SSV-II/results/g2-form-factor-results.md @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/papers/SSV-II/results/g2-form-factor-results.md
- papers/SSV-II/results/order-parameter/b3-chirality-from-spin-framing.md @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/papers/SSV-II/results/order-parameter/b3-chirality-from-spin-framing.md
- papers/SSV-II/results/reconnection/cap-observable-convergence.md @ d22eefe — https://github.com/StigNorland/SVT/blob/d22eefe/papers/SSV-II/results/reconnection/cap-observable-convergence.md
- papers/SSV-II/results/reconnection/validation-refinement-sweeps-reconnection-2026-05-28 @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/papers/SSV-II/results/reconnection/validation-refinement-sweeps-reconnection-2026-05-28.md
- papers/SSV-II/results/topological-vertex-exploration.md @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/papers/SSV-II/results/topological-vertex-explor md
- papers/SSV-IV/results/gw-pol-issue129.md @ cda2581 — https://github.com/StigNorland/SVT/blob/cda2581/papers/SSV-IV/results/gw-pol-issue129.md
- papers/SSV-IV/results/issue-119-falsification-and-bath-candidate.md @ 61bf640 — https://github.com/StigNorland/SVT/blob/61bf640/papers/SSV-IV/results/issue-119-falsification-a md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-II/CHANGELOG.md, and the full history is in the repository's commits.